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A low frequency oscillation suppression method for grid-connected DFIG with virtual inertia

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ABSTRACT

This paper addresses the problem that grid-connected doubly-fed induction generators (DFIG) with virtual inertia control cannot simultaneously take into account both rotor angle stability and system frequency regulation. Based on the dynamic energy curvature, a method of suppressing low frequency oscillation that considers the effect of frequency regulation is proposed. First, according to the relationship between the system stability and the dynamic energy of DFIG, the energy curvature model of the system is constructed. From that a detailed expression of the energy curvature is obtained, so that current compensation terms can be inversely deduced, and compensation branches can be constructed accordingly to improve the energy curvature during low frequency oscillation. Further, with the goal of optimizing both energy curvature and frequency regulation, a branch parameter optimization scheme is established. This scheme places equal emphasis on oscillation suppression and frequency support in order to maximize the capability of DFIG in both oscillation suppression and frequency regulation. Finally, the proposed method is verified through simulation tests performed in RTDS according to the parameters of a real grid-connected wind farm. The results show that, by ensuring the steady-state frequency deviation and reducing the maximum frequency deviation, this method can effectively increase the energy curvature, quickly suppress the low frequency oscillation in various disturbance cases, and simultaneously improve the frequency support capability of DFIG.

1. Introduction

In recent years, as represented by wind turbines, more and more equipment has been connected to the grid through power electronic inverters. With the introduction of grid-friendly guidelines for the connection of wind turbines in various countries [1], it is urgent to incorporate virtual inertia controls to wind turbines or wind farms so as to improve their grid-connected performance. However, when a wind turbine reaches a certain inertia threshold, its dynamic characteristics become coupled with the system's own characteristics. This will increase the influence of the synchronous generators on the movement of the rotor, and it may lead to intensified low frequency oscillation. Therefore, it is of great significance to explore measures that can simultaneously ensure stability of rotor angle and frequency in grid-connected wind power systems with virtual inertia control.

From a theoretical perspective, different types of analysis can be used to suppress low frequency oscillation. These include traditional control, optimal control, and adaptive control. Traditional control inputs the system's oscillation power, frequency, speed, and other feedback signals into the excitation voltage reference value in order to adjust the active power output [2]. On this basis, reference [3] introduced frequency feedback control to the MPPT control, so as to improve the system damping under low frequency oscillation through the optimized power point tracking control. Reference [4] improved the traditional PD control, so that it is more effective in suppressing low frequency oscillation. These traditional damping control methods do not include the now commonly used PLL control structure in the calculation and setting of parameters. As changes unfold, there is the possibility of worsening the vibration of the shaft, causing system resonance. Optimal control applies a variety of algorithms based on deterministic models or probabilistic small-signal models to damp multiple oscillation types by adjusting wind turbine damping controller parameters and wind farm coordinated controller parameters [5,6]. The accuracy of the optimal control method

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Nomenclature

U_{sd0}, I_{sd0}, I_{sq0}	DFIG terminal d-axis voltage, d-axis current and q-axis current at the equilibrium point
U_s, I_s, θ_s	DFIG terminal voltage, current and phase angle
U_{pcc}, θ_{pcc}	Voltage and phase angle at PCC
Z_L, θ_L	Equivalent impedance and angle of the outlet line
φ	DFIG power factor angle.
K_p, K_i	PI parameters of Power outer loop control
K_{p_pll}, K_{i_pll}	PI parameters of PLL.
K_{VI}	Differential coefficient of virtual inertia control
ω, γ	Radian frequency and attenuation coefficient of the low-frequency oscillation
Ω, Ω_U, Ω_I	Amplitude of the oscillation of phase angle of PCC, DFIG terminal voltage and current
H, D	Equivalent inertia time constant and damping coefficient of synchronous generator
α_i	Droop coefficient of the ith generator.
τ_i	Time constant of the governor of the ith generator.
K_{DFIG}	Capacity proportion of DFIG.
ω_s	System frequency.
$G_{DFIG}(s)$	Response function of DFIG primary frequency modulation

relies on the accuracy of the models used, and the algorithms' ability to solve high dimensionality and other complex problems. To achieve adaptive damping control, advanced methods for system analysis must be adopted. This means using wide-area measurement signals (WAMS) to coordinate and optimize the parameters of the synchronous generator and the wide-area damping controller of the wind farm [7]. Since this relies on FACTS equipment and HVDC transmission systems, the investment required is quite large, and the wide-area information that can be fed back by the phasor measurement unit (PMU) is limited [8], so the choice of its measurement signal still needs to be studied in depth. From the aforementioned research we can see that the existing methods present a conflict between suppressing low frequency oscillation and processing complex measurement data, so there is not a strong enough basis for selecting the measurement signal.

Besides, many references put forward methods to improve the virtual inertia control of DFIG. In reference [9], the frequency response characteristic of inverter control was improved by introducing virtual inertia to the voltage control link and current control link. Reference [10] designed an inertia controller containing the feedback of multiple variables, which can support the frequency when there is a long-term power shortage in the system. However, these methods do not consider that, while the system inertia is improved, the joint effect of PLL control and virtual inertia control may worsen the stability level of the system power angle. The issues of rotor angle stability and system frequency modulation studied in this paper are based on the classical virtual inertia control structure. As stated in reference [11], the classical virtual inertia control will magnify the amplitude of low frequency oscillation, thus it may worsen the small-disturbance stability of the system.

In view of the above problems, this paper uses the dynamic energy function of grid-connected DFIG with virtual inertia control to propose a branch compensation scheme that takes into account both oscillation suppression and frequency regulation. First, based on the relationship between the dynamic energy of DFIG and the system stability, the energy curvature model of the system is constructed. On this basis, a detailed expression of the energy curvature is obtained, thus the current compensation terms used to increase the energy curvature are inversely deduced. Then, supplementary branches are constructed to compensate the negative energy curvature terms. Further, with the goal of simultaneously optimizing the energy curvature and frequency regulation, a branch parameter optimization scheme with equal emphasis on oscillation suppression and frequency regulation is established, so that the oscillation suppression and frequency support capabilities of DFIG are maximized. Finally, combined with a real grid-connected wind farm, a simulation model is established in RTDS and multiple scenarios are set for verification. The results show the accuracy and effectiveness of the proposed method.

2. Design of low frequency oscillation suppression branches based on the energy curvature model

2.1. DFIG terminal energy curvature model

According to the terminal energy characteristic of grid-connected DFIG under low frequency oscillation [12], the concavity of the aperiodic component of DFIG terminal energy reflects the oscillatory stability of power system. Thus this paper defines the opposite number of the second-order derivative of terminal energy function ΔW^A as the energy curvature ΔE. It characterizes the stability trend of the total energy of system. The definition of ΔE is:

$$\Delta E = -\frac{d^2\Delta W^A}{dt^2}$$
$$\Delta P = -\frac{d\Delta W^A}{dt} \tag{1}$$

In (1), ΔW^A is equivalent to the original dynamic energy function of the generator; ΔE is equivalent to the 2nd-order derivative of the energy function. ΔP is equivalent to the 1st-order derivative of the energy function. When $\Delta E < 0$ and $\Delta P > 0$ (i.e. the original function decreases monotonically and the energy function concaves downward), the energy released by the system per unit time will first decrease to the valley, and then keep increasing until it surpasses the dissipation limit, and the system will become unstable. When $\Delta E > 0$ and $\Delta P < 0$ (i.e. the original function increases monotonically and the energy function convexes upward), the energy released by the system per unit time will first increase to the peak, and then keep decreasing until the system becomes stable. The dissipation limit refers to the boundary of the energy a system can release. When the energy released by the system goes beyond the energy boundary, the system will become unstable. The superscript A represents extracting the aperiodic component of the variable, and it is omitted in the following formulas.

When low frequency oscillation occurs in the system, the aperiodic component of DFIG terminal energy [13] is:

$$\Delta W_{DFIG} = \int_t^{t+\frac{2\pi}{\omega}} \Delta P d\Delta\theta_{pll} + \Delta i_d d\Delta u_q - \Delta i_q d\Delta u_d \tag{2}$$

The increments of current and active power in (2) can be divided into the phase angle swing components and excitation control effect components:

$$\begin{aligned}\Delta W_{DFIG} &= \int_t^{t+\frac{2\pi}{\omega}} \Delta P_e d\Delta\theta_{pll} + \Delta i_{sd} d\Delta u_{sq} - \Delta i_{sq} d\Delta u_{sd} \\ &\int_t^{t+\frac{2\pi}{\omega}} \Delta P^c_e d\Delta\theta_{pll} + \Delta i^c_{sd} d\Delta u_{sq} - \Delta i^c_{sq} d\Delta u_{sd} \\ &= \Delta W_s + \Delta W_c\end{aligned} \tag{3}$$

According to the contributing factor, the energy curvature in (3) can be divided into two parts: the energy curvature generated by the swing

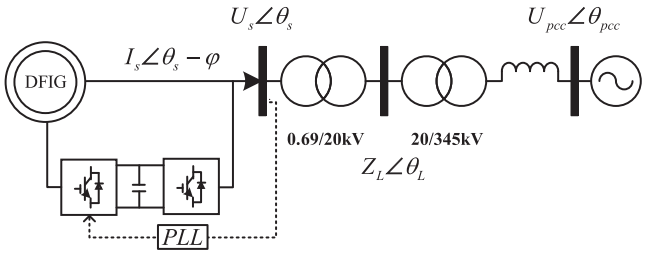


Fig. 1. DFIG-integrated power system.

of phase angle ΔE_S and the energy curvature generated by the excitation control effect ΔE_C.

$$\begin{cases} \Delta E_S = -\dfrac{d^2\Delta W_s}{dt^2} = \dfrac{d}{dt}\left(\Delta i_{sq}\Delta\dot{u}_{sd} - \Delta i_{sd}\Delta\dot{u}_{sq} - \Delta P_e\Delta\dot{\theta}_{pll}\right)\Big|_t^{t+\frac{2\pi}{\omega}} \\ \Delta E_C = -\dfrac{d^2\Delta W_c}{dt^2} = \dfrac{d}{dt}\left(\Delta i^c_{sq}\Delta\dot{u}_{sd} - \Delta i^c_{sd}\Delta\dot{u}_{sq} - \Delta P^c_e\Delta\dot{\theta}_{pll}\right)\Big|_t^{t+\frac{2\pi}{\omega}} \end{cases} \tag{4}$$

(1) Energy curvature generated by the swing of phase angle:

When low frequency oscillation occurs in the DFIG-integrated system, the increment of active power can be expressed as:

$$\Delta P_e + \Delta P^c_e = U_{sd0}\Delta i_{sd} + I_{sd0}\Delta u_{sd} + I_{sq0}\Delta u_{sq} + U_{sd0}\Delta i^c_{sd} \tag{5}$$

where U_{sd0}, I_{sd0} and I_{sq0} are DFIG terminal d-axis voltage, d-axis current and q-axis current at the equilibrium point.

Under low frequency oscillation, the three-phase phase lock loops (PLL) can be linearized as follows [13]:

$$\frac{d\Delta\theta_{pll}}{dt} = K_{p_pll}\Delta u_{sq} + K_{i_pll}\int \Delta u_{sq}dt \tag{6}$$

where K_{p_pll} and K_{i_pll} are the proportion and integration coefficients of PLL. By applying (5) and (6) to (4), ΔE_s can be obtained:

$$\Delta E_S = \Delta E_{SP} + \Delta E_{SI} + \Delta E_{SL} \tag{7}$$

The energy curvature terms in (7) can be expressed with the perturbations of voltage and current as:

$$\begin{aligned} \Delta E_{SP} &= K_{p_pll}\frac{d}{dt}\left(\begin{matrix} -U_{sd0}\Delta i_{sd}\Delta u_{sq} - I_{sd0}\Delta u_{sd}\Delta u_{sq} \\ -I_{sq0}\Delta u_{sq}\Delta u_{sq} \end{matrix}\right)\Big|_t^{t+\frac{2\pi}{\omega}} \\ &= \Delta E_{SP1} + \Delta E_{SP2} + \Delta E_{SP3} \\ \Delta E_{SI} &= K_{i_pll}\frac{d}{dt}\left(\begin{matrix} -U_{sd0}\Delta i_{sd}\int \Delta u_{sq}dt - I_{sd0}\Delta u_{sd}\int \Delta u_{sq}dt \\ -I_{sq0}\Delta u_{sq}\int \Delta u_{sq}dt \end{matrix}\right)\Big|_t^{t+\frac{2\pi}{\omega}} \\ &= \Delta E_{SI1} + \Delta E_{SI2} + \Delta E_{SI3} \\ \Delta E_{SL} &= \frac{d}{dt}(\Delta i_{sq}\dot{\Delta u}_{sd} - \Delta i_{sd}\dot{\Delta u}_{sq})\Big|_t^{t+\frac{2\pi}{\omega}} = \Delta E_{SL1} + \Delta E_{SL2} \end{aligned} \tag{8}$$

where ΔE_{SP} is the phase angle swing energy curvature dominated by the PLL proportion link. ΔE_{SI} is the phase angle swing energy curvature dominated by the PLL integration link. ΔE_{SL} is the phase angle swing energy curvature caused by the coupling between d axis and q axis.

(2) Energy curvature generated by the excitation control effect:

The induction increment of DFIG stator current [14] is:

$$\Delta i^c_{sdq} = -\frac{L_m}{L_s}\Delta i_{rdq} \tag{9}$$

By applying (5) and (9) to (4), the energy curvature generated by the excitation control effect can be obtained:

$$\Delta E_C = \Delta E_{CP} + \Delta E_{CI} + \Delta E_{CL} \tag{10}$$

The energy curvature terms in (10) can be expressed with the perturbations of voltage and current as:

$$\begin{aligned} \Delta E_{CP} &= K_{p_pll}\frac{L_m}{L_s}U_{sd0}\frac{d}{dt}(\Delta i_{rd}\Delta u_{sq})\Big|_t^{t+\frac{2\pi}{\omega}} \\ \Delta E_{CI} &= K_{i_pll}\frac{L_m}{L_s}U_{sd0}\frac{d}{dt}(\Delta i_{rd}\int \Delta u_{sq}dt)\Big|_t^{t+\frac{2\pi}{\omega}} \\ \Delta E_{CL} &= \frac{L_m}{L_s}\frac{d}{dt}(\Delta i_{rd}\dot{\Delta u}_{sq} - \Delta i_{rq}\dot{\Delta u}_{sd})\Big|_t^{t+\frac{2\pi}{\omega}} = \Delta E_{CL1} + \Delta E_{CL2} \end{aligned} \tag{11}$$

where ΔE_{CP} is the excitation control energy curvature dominated by the PLL proportion link. ΔE_{CI} is the excitation control energy curvature dominated by the PLL integration link. ΔE_{CL} is the excitation control energy curvature caused by the coupling between d axis and q axis.

2.2. Calculation of DFIG terminal energy curvature

The structure of a DFIG-integrated power system is shown in Fig. 1. The wind turbine is connected to the 20 kV bus via the step-up transformer (0.69 kV/20 kV), and then integrated into the grid via the 20 kV transmission line and the step-up transformer (20 kV/345 kV). U_s is the DFIG terminal voltage. Z_L is the equivalent impedance of the outlet line. U_{pcc} is the voltage at PCC (point of common coupling). θ_s is the DFIG terminal phase angle. θ_{pcc} is the phase angle at PCC. I_s is the current on the outlet line. θ_L is the impedance angle of the transmission line. φ is the power factor angle.

Apply dq transformation to the voltages and currents at different nodes and then linearize the results, so that the variables in (8)-(10) can be expressed with the increments of phase angles as:

$$\begin{cases} \Delta u_{sd} = -U_s\sin(\theta_{s.0} - \theta_{pll.0})(\Delta\theta_s - \Delta\theta_{pll}) \\ \Delta u_{sq} = U_s\cos(\theta_{s.0} - \theta_{pll.0})(\Delta\theta_s - \Delta\theta_{pll}) \end{cases}$$
$$\begin{cases} \Delta u_{pccd} = -U_{pcc}\sin(\theta_{pcc.0} - \theta_{pll.0})(\Delta\theta_{pcc} - \Delta\theta_{pll}) \\ \Delta u_{pccq} = U_{pcc}\cos(\theta_{pcc.0} - \theta_{pll.0})(\Delta\theta_{pcc} - \Delta\theta_{pll}) \end{cases} \tag{12}$$
$$\begin{cases} \Delta i_{sd} = -I_s\sin(\theta_{s.0} - \theta_{pll.0} - \varphi)(\Delta\theta_s - \Delta\theta_{pll} - \Delta\varphi) \\ \Delta i_{sq} = I_s\cos(\theta_{s.0} - \theta_{pll.0} - \varphi)(\Delta\theta_s - \Delta\theta_{pll} - \Delta\varphi) \end{cases}$$

When low frequency oscillation occurs in the system, the voltage at PCC and the DFIG terminal voltage are disturbed. According to Fig. 1, the relationship between the two voltages is:

$$U_s\angle\theta_s = U_{pcc}\angle\theta_{pcc} + Z_L\angle\theta_L\cdot I_s\angle\theta_s - \varphi \tag{13}$$

Apply dq transformation to (13), and then apply (12) and (13), so that:

$$\frac{\Delta\theta_s - \Delta\theta_{pll}}{\Delta\theta_{pcc} - \Delta\theta_{pll}} = \frac{U_{pcc}\sin(\theta_Z + \theta_{s.0} - \varphi - \theta_{pcc.0})}{U_s\sin(\theta_Z - \varphi)} \tag{14}$$

According to (6) and (14), $\Delta\theta_{pll}$, $\Delta\theta_s$ and $\Delta\theta_{pcc}$ have the same frequency, and they can all be written in the following form (see Appendix

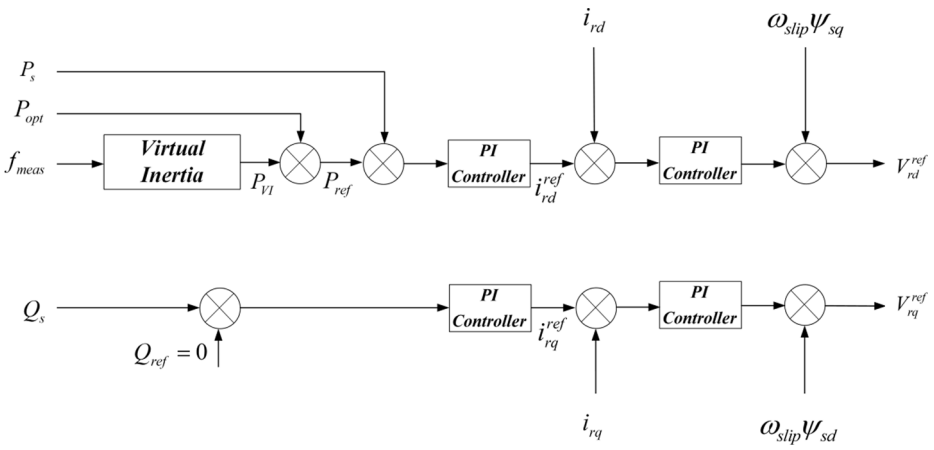


Fig. 2. Control diagram of DFIG rotor-side converter.

A for the derivation process):

$$\begin{cases} \Delta\theta_{pcc}-\Delta\theta_{pll} = \Omega e^{\gamma t}\sin(\omega t+\theta) \\ \Delta\theta_s-\Delta\theta_{pll} = \Omega_U e^{\gamma t}\sin(\omega t+\theta) \\ \Delta\theta_s-\Delta\theta_{pll}-\Delta\varphi = \Omega_I e^{\gamma t}\sin(\omega t+\theta) \end{cases} \tag{15}$$

By applying (12) and (15) to (8), the expressions of the energy curvature terms in ΔE_S can be obtained:

$$\begin{aligned} \Delta E_{SP1} &= K_{p_pll}\Omega_U\Omega_I U_{sd0}U_s I_s sin(\theta_{s.0}-\theta_{pll.0}-\varphi) \\ &\quad \hat{\mathrm{A}}\cdot cos(\theta_{s.0}-\theta_{pll.0})\gamma e^{2\gamma t}\big|_t^{t+\frac{2\pi}{\omega}} \\ \Delta E_{SP2} &= K_{p_pll}\Omega_U^2 I_{sd0}U_s^2 sin(\theta_{s.0}-\theta_{pll.0}) \\ &\quad \hat{\mathrm{A}}\cdot cos(\theta_{s.0}-\theta_{pll.0})\gamma e^{2\gamma t}\big|_t^{t+\frac{2\pi}{\omega}} \\ \Delta E_{SP3} &= -K_{p_pll}\Omega_U^2 U_s^2 I_{sq0}\cos^2(\theta_{s.0}-\theta_{pll.0})\gamma e^{2\gamma t}\big|_t^{t+\frac{2\pi}{\omega}} \\ \Delta E_{SI1} &= \frac{K_{i_pll}\gamma}{\gamma^2+\omega^2}\Omega_U\Omega_I U_s U_{sd0}I_s sin(\theta_{s.0}-\theta_{pll.0}-\varphi) \\ &\quad \cdot cos(\theta_{s.0}-\theta_{pll.0})\gamma e^{2\gamma t}\big|_t^{t+\frac{2\pi}{\omega}} \\ \Delta E_{SI2} &= \frac{K_{i_pll}\gamma}{\gamma^2+\omega^2}\Omega_U^2 U_s^2 I_{sd0} sin(\theta_{s.0}-\theta_{pll.0}) \\ &\quad \cdot cos(\theta_{s.0}-\theta_{pll.0})\gamma e^{2\gamma t}\big|_t^{t+\frac{2\pi}{\omega}} \\ \Delta E_{SI3} &= -\frac{K_{i_pll}\gamma}{\gamma^2+\omega^2}\Omega_U^2 U_s^2 I_{sq0}\cos^2(\theta_{s.0}-\theta_{pll.0})\gamma e^{2\gamma t}\big|_t^{t+\frac{2\pi}{\omega}} \\ \Delta E_{SL1} &= \Omega_U\Omega_I U_s I_s sin(\theta_{s.0}-\theta_{pll.0}-\varphi) \\ &\quad \cdot cos(\theta_{s.0}-\theta_{pll.0})\gamma e^{2\gamma t}\big|_t^{t+\frac{2\pi}{\omega}} \\ \Delta E_{SL2} &= -\Omega_U\Omega_I U_s I_s sin(\theta_{s.0}-\theta_{pll.0}) \\ &\quad \cdot cos(\theta_{s.0}-\theta_{pll.0}-\varphi)\gamma e^{2\gamma t}\big|_t^{t+\frac{2\pi}{\omega}} \end{aligned} \tag{16}$$

When calculating the energy corresponding to DFIG rotor excitation control response, the bandwidth of the current inner loop can be neglected in the electromechanical time scale, and only the outer loop power control is retained [15]. The control diagram of DFIG rotor-side converter with virtual inertia is shown in Fig. 2.

According to the division of the time scale in reference [16], the time scale of the DC regulation within the DFIG converter is much smaller than the electromechanical time scale of low frequency oscillation. Hence, the effect of the DC regulation within the DFIG converter on the system stability in the electromechanical time scale is very small, which can be neglected.

According to Fig. 2, the rotor current response can be written as:

$$\begin{cases} \Delta i_{rd} = K_p(\Delta P_{ref}-\Delta P)+K_i\int(\Delta P_{ref}-\Delta P)dt \\ \Delta i_{rq} = K_p(\Delta Q_{ref}-\Delta Q)+K_i\int(\Delta Q_{ref}-\Delta Q)dt \end{cases} \tag{17}$$

where K_p and K_i are the PI parameters of the active power and reactive power outer loop. Since P_{ref0} provided by the rotor speed control remains unchanged, ΔP_{ref}, ΔQ_{ref} and ΔQ can be expressed as:

$$\begin{cases} \Delta P_{ref} = \Delta P_{VI} = -K_{VI}\Delta\dot{f}_{pll} = -K_{VI}\Delta\ddot{\theta}_{pll} \\ \Delta Q_{ref} = 0 \\ \Delta Q = I_{sd0}\Delta u_{sq}-U_{sd0}\Delta i_{sq}-I_{sq0}\Delta u_{sd} \end{cases} \tag{18}$$

Applying (6), (12), (15) and (18) to (17) yields (see Appendix B for the derivation process):

$$\begin{cases} \Delta i_{rd} = A_d e^{\gamma t}\sin(\omega t+\theta)+B_d e^{\gamma t}\cos(\omega t+\theta) \\ \Delta i_{rq} = A_q e^{\gamma t}\sin(\omega t+\theta)+B_q e^{\gamma t}\cos(\omega t+\theta) \end{cases} \tag{19}$$

By applying (12), (15) and (19) to (11), the expressions of the energy curvature terms in ΔE_C can be obtained:

$$\begin{aligned} \Delta E_{CP} &= K_{p_pll}\frac{L_m}{L_s}\Omega_U A_d U_s U_{sd0} cos(\theta_{s.0}-\theta_{pll.0})\gamma e^{2\gamma t}\big|_t^{t+\frac{2\pi}{\omega}} \\ \Delta E_{CI} &= K_{i_pll}\frac{L_m}{L_s}\frac{A_d\gamma-B_d\omega}{\gamma^2+\omega^2}\Omega_U U_s U_{sd0} cos(\theta_{s.0}-\theta_{pll.0}) \\ &\quad \cdot\gamma e^{2\gamma t}\big|_t^{t+\frac{2\pi}{\omega}} \\ \Delta E_{CL1} &= \frac{L_m}{L_s}(A_d\gamma+B_d\omega)\Omega_U U_s cos(\theta_{s.0}-\theta_{pll.0})\gamma e^{2\gamma t}\big|_t^{t+\frac{2\pi}{\omega}} \\ \Delta E_{CL2} &= \frac{L_m}{L_s}(A_q\gamma+B_q\omega)\Omega_U U_s sin(\theta_{s.0}-\theta_{pll.0})\gamma e^{2\gamma t}\big|_t^{t+\frac{2\pi}{\omega}} \end{aligned} \tag{20}$$

The coefficients in the energy curvature terms in (20) are shown below:

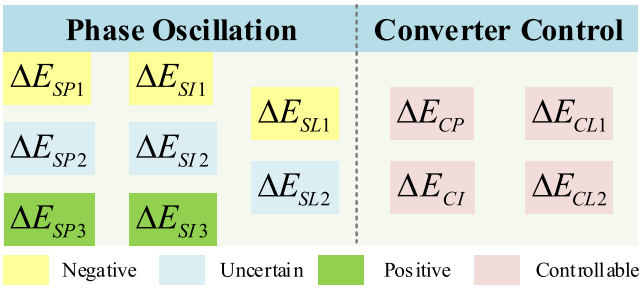


Fig. 3. Signs of different energy curvature terms.

$$\begin{cases} A_d = \begin{bmatrix} \Omega_I U_{sd0} I_s \sin(\theta_{s.0} - \theta_{pll.0} + \varphi) \\ +\Omega_U U_s I_{sd0} \sin(\theta_{s.0} - \theta_{pll.0}) \\ -\Omega_U U_s I_{sq0} \cos(\theta_{s.0} - \theta_{pll.0}) \end{bmatrix} \left(K_p + \frac{K_i \gamma}{\gamma^2 + \omega^2} \right) \\ -\Omega_U U_s \cos(\theta_{s.0} - \theta_{pll.0}) K_{VI} \begin{pmatrix} K_{p_pll} K_p \gamma + K_{i_pll} K_p \\ +K_{p_pll} K_i + \frac{K_{i_pll} K_i \gamma}{\gamma^2 + \omega^2} \end{pmatrix} \\ A_q = \begin{bmatrix} \Omega_U U_s I_{sq0} \cos(\theta_{s.0} - \theta_{pll.0}) \\ -\Omega_I U_{sd0} I_s \sin(\theta_{s.0} - \theta_{pll.0} + \varphi) \\ -\Omega_U U_s I_{sd0} \sin(\theta_{s.0} - \theta_{pll.0}) \end{bmatrix} \frac{K_i \omega}{\gamma^2 + \omega^2} \end{cases} \tag{21}$$

$$\begin{cases} B_d = -\begin{bmatrix} \Omega_I U_{sd0} I_s \sin(\theta_{s.0} - \theta_{pll.0} + \varphi) \\ -\Omega_U U_s I_{sd0} \sin(\theta_{s.0} - \theta_{pll.0}) \\ -\Omega_U U_s I_{sq0} \cos(\theta_{s.0} - \theta_{pll.0}) \end{bmatrix} \frac{K_i \omega}{\gamma^2 + \omega^2} \\ -\Omega_U U_s \cos(\theta_{s.0} - \theta_{pll.0}) K_{VI} \begin{pmatrix} K_{p_pll} K_p \omega \\ -\frac{K_{i_pll} K_i \omega}{\gamma^2 + \omega^2} \end{pmatrix} \\ B_q = \begin{bmatrix} \Omega_U U_s I_{sd0} \cos(\theta_{s.0} - \theta_{pll.0}) \\ +\Omega_U U_s I_{sq0} \sin(\theta_{s.0} - \theta_{pll.0}) \\ -\Omega_I U_{sd0} I_s \cos(\theta_{s.0} - \theta_{pll.0} + \varphi) \end{bmatrix} \frac{K_i \omega}{\gamma^2 + \omega^2} \end{cases}$$

2.3. Branches design based on negative energy curvature compensation

According to the above analysis, the equivalent model of converter station in flexible DC grid can be built, as shown in Fig. 3.

According to the relationship between the dynamic energy of DFIG and the system stability, the energy curvature model of the system is established. On this basis, the detailed expression of energy curvature after the system is disturbed is derived from the flow path of the oscillation components. And then, the compensation term which can increase the energy curvature is obtained from reverse derivation, and a compensation branch is constructed to compensate the negative energy curvature term. After the compensation, the total energy curvature of the system can be set at the stable state before the oscillation occurs, so that the oscillation can be suppressed in Ref. [14]. Thus, this paper calculates the energy curvature of DFIG under the present control structure, and analyzes the signs (positive or negative) of different energy curvature terms, so that the energy curvature can be increased by adding supplementary compensatation branches to the coresponding energy curvature terms.

In the steady state, the stability margin coefficient of power system $\sin(\theta_{s.0} - \theta_{pcc.0}) > 0$. The power factor angle of DFIG $\varphi > 0$. The impedance angle of DFIG outlet line is $\theta_Z \approx 90^\circ$. The operating condition of PLL is ideal, with $\theta_{s.0} - \theta_{pll.0} \approx 0$. Applying the above descriptions to (15) yields that $\Omega_U > 0$ and $\Omega_I > 0$. Then, by combining (16) and (20), the signs of different energy curvature terms can be obtained, as shown in Fig. 3.

According to Fig. 3, energy curvature terms ΔE_{SP1}, ΔE_{SI1}, and ΔE_{SL1} are negative. They have negative damping effects on the oscillation and make against the stability of system, thus they need to be compensated. The signs of ΔE_{SP2}, ΔE_{SI2} are ΔE_{SL2} are uncertain, depending on $\sin(\theta_{s.0}$-$\theta_{pll.0})$. But since the difference between the phase angles of PLL and bus voltage at the beginning of oscillation is very small, the uncertain terms can be neglected. ΔE_{CP}, ΔE_{CI}, ΔE_{CL1} and ΔE_{CL2} are controlled energy curvature terms, with A_d, B_d, A_q and B_q as controllable parameters affected by the control parameters of the outer loop and virtual inertia. In order to guarantee the reliability of fundamental frequency control of the converter, this paper does not change the original control parameters, but changes the expressions of the controllable parameters by adding a compensation branch to the rotor current. In this way, the curvature of DFIG terminal energy can be changed. Hence, the object of compensation includes ΔE_{SP1}, ΔE_{SI1}, ΔE_{SL1} and the negative curvature terms in ΔE_{CP}, ΔE_{CI}, ΔE_{CL1} and ΔE_{CL2}.

As can be seen from (8), ΔE_{SP1}, ΔE_{SI1} and ΔE_{SL1} are all negative curvature terms generated by the coupling between the d-axis current swing component and q-axis voltage swing component in different control links. while ΔE_{CP}, ΔE_{CI} and ΔE_{CL1} are energy curvature terms generated by the coupling between the d-axis current control component and q-axis voltage swing component. The two groups of curvature terms are calculated in the same manner, thus the negative curvature terms can be compensated by adding a compensation term to the swing component of the rotor current. ΔE_{SP1} and ΔE_{CP} can be expressed with the perturbations of voltage and current as (in the per unit system):

$$\Delta E_{SP1} = K_{p_pll} U_{sd0} \frac{d}{dt} \left(-\Delta i_{sd} \Delta u_{sq} \right) \Big|_t^{t+\frac{2\pi}{\omega}} \tag{22}$$

$$\Delta E_{CP} = K_{p_pll} U_{sd0} \frac{L_m}{L_s} \frac{d}{dt} \left(\Delta i_{rd} \Delta u_{sq} \right) \Big|_t^{t+\frac{2\pi}{\omega}} \tag{23}$$

For the two energy curvature terms in (22) and (23), if Δi_{rd} is increased by $\frac{L_s}{L_m} \Delta i_{sd}$, the positive energy curvature generated by the increase will offset ΔE_{SP1}. Thus, the 1st current compensation branch can be constructed:

$$\Delta i_{rd}^{(1)} = \frac{L_s}{L_m} \Delta i_{sd} \tag{24}$$

The same compensation principle applies to ΔE_{SL1} and ΔE_{CL1} in (11).

The variation of parameters will cause the related energy curvature terms to change, thus ΔE_c is re-divided by the curvature terms corresponding to different parameters:

$$\begin{aligned} \Delta E_c = \frac{U_s \Omega_s L_m}{L_s} \Bigg[& A_d \left(\frac{U_{sd0} K_{p_pll}}{\gamma^2 + \omega^2} + \gamma \right) \\ & + B_d \left(\omega - \frac{U_{sd0} K_{i_pll} \omega}{\gamma^2 + \omega^2} \right) \Bigg] e^{2\gamma t} \Big|_t^{t+\frac{2\pi}{\omega}} \\ & \cdot cos(\theta_{s.0} - \theta_{pll.0}) \\ & + (A_q \gamma + B_q \omega) sin(\theta_{s.0} - \theta_{pll.0}) \\ = & A_d \Delta E_{Ad} + B_d \Delta E_{Bd} + A_q \Delta E_{Aq} + B_q \Delta E_{Bq} \end{aligned} \tag{25}$$

When low frequency oscillation occurs, the following inequalities are satisfied:

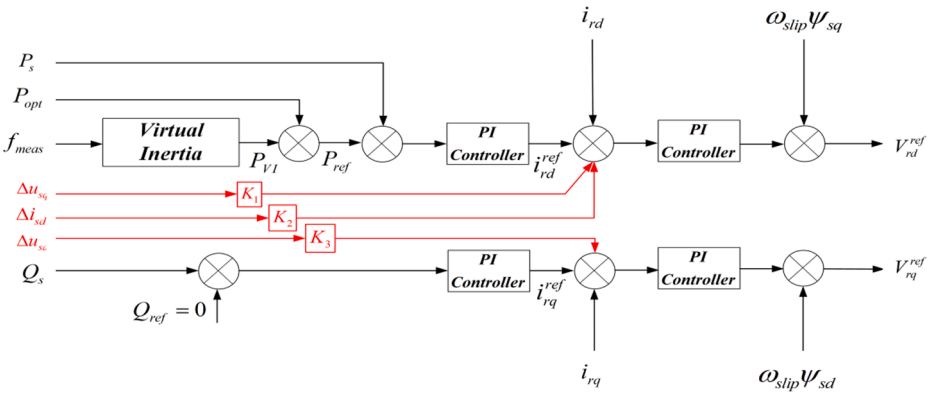


Fig. 4. Control diagram of DFIG rotor-side converter with compensation branches.

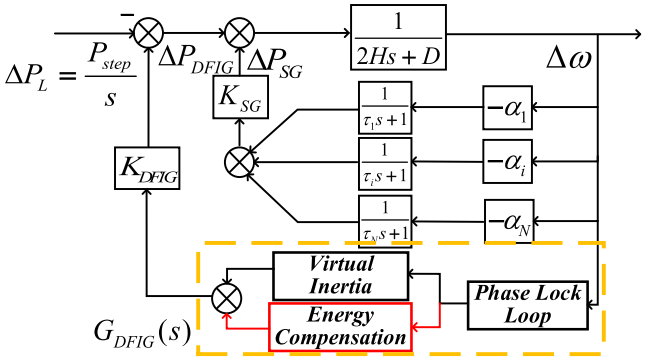


Fig. 5. Diagram of DFIG and synchronous generators jointly participating in frequency modulation.

$$\begin{cases} \omega \gg \gamma \\ U_{sd0}K_{p_pll} + U_{sd0}K_{i_pll}\dfrac{\gamma}{\gamma^2+\omega^2} + \gamma > 0 \end{cases} \tag{26}$$

In (25), the curvature terms corresponding to different parameters are jointly affected by the operation state and the oscillation mode, etc.. $\Delta E_{Ad} > 0$. The magnitude of ΔE_{Aq} and is very small. The sign of ΔE_{Bd} is uncertain, but $\Delta E_{Bd} \ll \Delta E_{Ad}$ in the low frequency band and $\Delta E_{Aq} \ll \Delta E_{Bq}$ because $\omega \gg \gamma$. Thus, compared with the negative term in $A_d\Delta E_{Ad}$ and $B_q\Delta E_{Bq}$, the negative term in $B_d\Delta E_{Bd}$ and $A_q\Delta E_{Aq}$ is negligible and need not be compensated. According to (19), $B_d\Delta E_{Bd}$ and $A_q\Delta E_{Aq}$ is not affected when compensating the negative terms in $A_d\Delta E_{Aq}$ and $B_q\Delta E_{Bq}$.

According to (19), the negative coefficient in A_d is generated by the coupling between PLL and virtual inertia control. The negative coefficient in B_q which has the different sign as $\sin(\theta_{s.0} - \theta_{pll.0})$ is generated by the integration of d-axis voltage. Referring to the derivation process in Appendix B, the sources of these coefficients can be found. Thus, by compensating the corresponding energy terms, the negative curvature caused by the excitation control effect can be offset. The 2rd and 3th current compensation branches are:

$$\Delta i_{rd}^{(2)} = K_{VI}\left(K_pK_{i_pll} + K_iK_{p_pll}\right)\Delta u_{sq} + U_{sd0}\Delta i_{sd} \tag{27}$$

$$\Delta i_{rq}^{(3)} = -K_iI_{sq0}\int \Delta u_{sd}dt \tag{28}$$

The first two compensation branches compensate the d-axis rotor current, and the 3th compensation branch compensates the q-axis rotor current. The control diagram of rotor-side converter with compensation branches is shown in Fig. 4.

Combine the expressions of the compensation branches in (24), (27) and (39) with (11), and substitute the original rotor current with the sum of the compensation currents, so that the increase of energy curvature ΔE_{EC} caused by the supplementary compensation branches can be calculated:

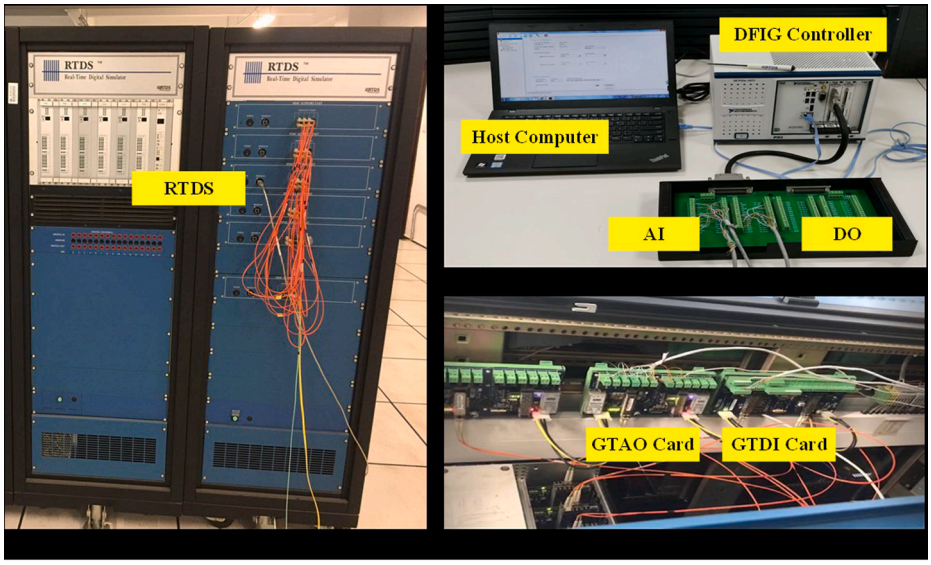


Fig. 6. Flowchart calculating optimal parameters.

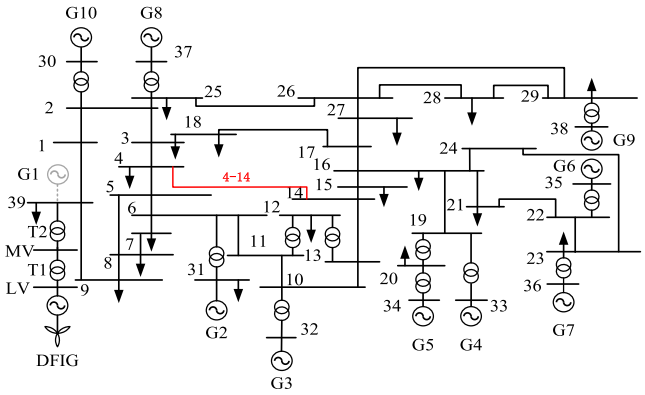


Fig. 7. Experimental platform.

$$\Delta E_{EC} = \left[\begin{array}{c} -(K_{p_pll}U_{sd0} + \dfrac{K_{i_pll}U_{sd0}\gamma}{\gamma^2+\omega^2} + 1) \\ \cdot\Omega_U\Omega_I U_s I_s sin(\theta_{s.0} - \theta_{pll.0} - \varphi) cos(\theta_{s.0} - \theta_{pll.0}) \\ -(K_{p_pll} + \dfrac{K_{i_pll}\gamma}{\gamma^2+\omega^2} + \dfrac{1}{U_{sd0}}) \\ \cdot\Omega_U^2 U_s^2 I_{sq0} \cos^2(\theta_{s.0} - \theta_{pll.0}) \\ + \dfrac{L_m K_{VI}(K_{i_pll}K_p + K_{p_pll}K_i)}{L_s} \cos^2(\theta_{s.0} - \theta_{pll.0}) \\ \cdot(U_{sd0}K_{p_pll} + U_{sd0}\dfrac{K_{i_pll}\gamma}{\gamma^2+\omega^2} + \gamma)\Omega_U^2 U_s^2 \\ + \dfrac{L_m I_{sq0} K_i(\gamma^2-\omega^2)}{L_s(\gamma^2+\omega^2)} \Omega_U^2 U_s^2 \sin^2(\theta_{s.0} - \theta_{pll.0}) \\ \cdot\gamma e^{2\gamma t}\Big|_t^{t+\frac{2\pi}{\omega}} \end{array} \right] \quad (29)$$

By comparing the compensated energy curvature in (29) with the original negative energy curvature in (16) and (20), it can be seen that the two offset each other. The larger the diverging trend of the original oscillation γ is, the more positive energy curvature is compensated. Therefore, the compensation branches can effectively improve the stability of low frequency oscillation in DFIG-integrated system. These branches modify the power output by compensating part of the output energy of virtual inertia control, while the virtual inertia participates in the frequency control of power system and provides inertia support by changing the active power output of DFIG through the supplementary power control link [17]. Therefore, these branches will change the time-domain and frequency-domain characteristics of DFIG inertia, thus affecting the frequency support capability of DFIG during the primary

Table 1
DFIG Parameters.

Parameter	Symbol	Value	Parameter	Symbol	Value
Rated power	P_M	1.5 MW	Magnetic reactance	X_M	5.43Wb
Rated frequency	f	60 Hz	Rotor inertia	J	75Kg·m2
Stator voltage	U_s	0.69KV	Number of pole pairs	n_p	2
Stator resistance	R_s	0.01p.u	Rotor resistance	R_r	0.01p.u.
Stator inductance	L_s	0.1p.u.	Rotor reactance	X_r	0.1p.u.

The main parameters of DFIG rotor-side converter are shown in Table 2.

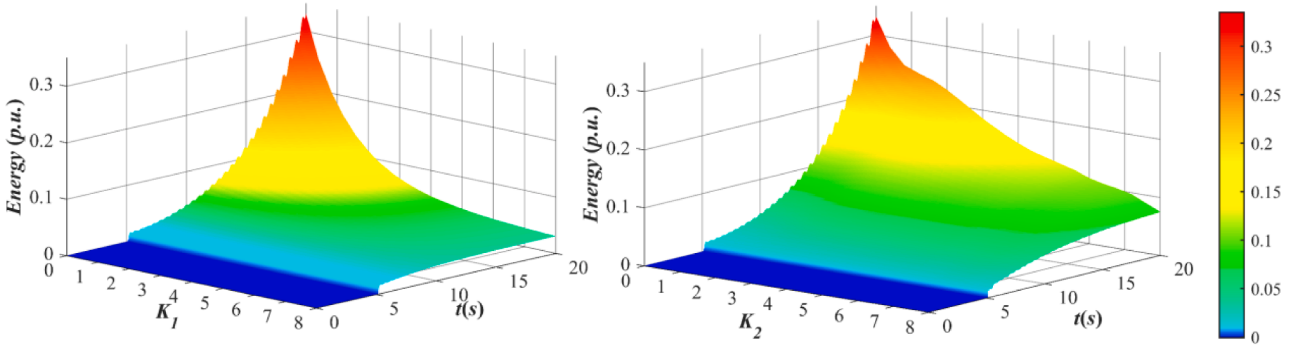
Table 2
Parameters of DFIG Rotor-Side Converter.

Parameter	Symbol	Value	Parameter	Symbol	Value
Rated line voltage	U_N	0.69 kV	Coefficient of virtual inertia	K_{VI}	0.2
Proportion coefficient of outer loop	Kp	4	Integration coefficient of outer loop	K_i	40
Proportion coefficient of PLL	K_{p_pll}	3	Integration coefficient of PLL	K_{i_pll}	10

The main parameters of the synchronous generators and test system are shown in Tables 3 and 4.

Table 3
Parameters of Synchronous Generators.

No. i	Capacity S/MVA	Time constant of inertia H/s	Damping coefficient D/p.u.	Time constant of governor T/s	Magnification of governor σ
02	700	4.329	1	2.283	29
03	800	4.475	1	2.283	29
04	800	3.575	1	2.283	29
05	300	4.333	1	2.283	29
06	800	4.35	1	2.283	29
07	700	3.777	1	2.283	29
08	700	3.471	1	2.283	29
09	1000	3.45	1	2.283	29
10	1000	4.2	1	2.283	29



(a) Branch 1 (b) Branch 2.

Fig. 8. DFIG-integrated 10-machine 39-bus system.

frequency modulation. Since different branches contribute differently to the compensation effect and frequency support effect, the control parameters of the supplementary energy branches should be set appropriately to balance the compensation effect and frequency support effect.

3. Parameter optimization considering oscillation suppression and frequency regulation

Based on the analysis in the last part of Section 2, by establishing indexes for the oscillation suppression capability and frequency support capability of the compensation branches and constructing the corresponding objective functions, the control parameters can be set through optimization of the multi-objective functions.

3.1. Oscillation suppression capability of compensation branches

It can be seen from (29) that, the compensation capability of different branches concerning the energy curvature are different. ΔE_{EC} in (29) can be written according to different branches as:

$$\Delta E_{EC1} = K_1 \left[\begin{array}{c} K_{p_pll} U_{sd0} \\ +\gamma\left(\frac{U_{sd0}K_{i_pll}}{\gamma^2+\omega^2}+1\right) \end{array} \right] U_s I_s \frac{L_m}{L_s}\Omega_U\Omega_I \cdot\sin(\theta_{pll.0}-\theta_{s.0}+\varphi)\cos(\theta_{s.0}-\theta_{pll.0})\gamma e^{2\gamma t}\Big|_t^{t+\frac{2\pi}{\omega}} \tag{30}$$

$$\Delta E_{EC2} = K_2\left[U_{sd0}K_{p_pll}+\gamma\left(\frac{U_{sd0}K_{i_pll}}{\gamma^2+\omega^2}+1\right)\right] \cdot\frac{L_m}{L_s}\Omega_U^2 U_s^2\cos^2(\theta_{s.0}-\theta_{pll.0})\gamma e^{2\gamma t}\Big|_t^{t+\frac{2\pi}{\omega}} \tag{31}$$

$$\Delta E_{EC3} = K_3\left[\frac{I_{sq0}K_i(\gamma^2-\omega^2)}{\gamma^2+\omega^2}\right]\frac{L_m}{L_s}\Omega_U^2 U_s^2 \cdot\sin^2(\theta_{s.0}-\theta_{pll.0})\gamma e^{2\gamma t}\Big|_t^{t+\frac{2\pi}{\omega}} \tag{32}$$

In (32), the coefficient contains a higher-order infinitesimal term $\sin^2(\theta_{s.0}-\theta_{pll.0})$, so the compensation capability of branch 3 is much lower than the other two branches. In the following evaulation of compensation capability, branch 3 is neglected.

The compensation capability of branch 1 and branch 2 is evaluated by the energy curvature each branch can compensate per unit time:

$$\eta_i = \frac{\Delta E_{ECi}}{\sum \Delta E_{ECi}/K_i} \tag{33}$$

where η_i reflects the capability of the ith branch in providing positive curvature (i = 1,2). When constructing the objective function of oscillation suppression, η_1 and η_2 are used as the indexes for the oscillation suppression capability of branch 1 and branch 2.

3.2. Frequency support capability of compensation branche

DFIG with its compensation branches participates in the primary frequency modulation of the system under the load impact, and this process is shown in Fig. 5. (see Fig. 6).

In Fig. 5, P_{step} is the power shortage. H is the equivalent inertia time constant of synchronous generator. D is the equivalent damping coefficient. α_i is the droop coefficient of the ith generator. τ_i is the time constant of the governor of the ith generator. K_{SG} is the capacity proportion of synchronous generator. K_{DFIG} is the capacity proportion of DFIG. $\Delta\omega$ is the perturbation of system frequency. $G_{DFIG}(s)$ is the response function of DFIG primary frequency modulation (see Appendix C for the derivation process).

According to Fig. 5, the energy compensation branches affect the frequency support capability mainly by changing the active power response of DFIG (branch 1 and branch 2). Branch 3 changes the structure of reactive power control, but the reactive power does not affect the dynamic process of primary frequency modulation. Thus branch 3 does not contribute to the frequency support capability.

When the secondary frequency modulation effect of AGC (automatic generation control) of synchronous generator is neglected, the frequency deviation of power system is strictly limited by: ① the maximum frequency deviation $\Delta\omega_{max}$. ② the steady-state frequency deviation $\Delta\omega_{st}$ [18]. After the DFIG-integrated system is impacted by the load with amplitude of P_{step}, the frequency response of the system is:

$$\Delta\omega = \frac{P_{step}}{s}\frac{1}{2Hs+D+K_{SG}\sum_i^N\frac{\alpha_i}{\tau_i s+1}+K_{DFIG}G_{DFIG}(s)} \tag{34}$$

Based on the final value theorem, the steady-state frequency deviation $\Delta\omega_{st}$ can be calculated according to (34):

$$\Delta\omega_{st} = \lim_{s\to 0} s\Delta\omega(s) = -\frac{P_{step}}{D+K_{SG}\sum\alpha_i} \tag{35}$$

According to (36), the compensation branches do not affect $\Delta\omega_{st}$, but $\Delta\omega_{max}$ is affected by parameters K_1 and K_2 of the compensation branches. Thus $\Delta\omega_{max}$ (K_1, K_2) is used as the index for the construction of the objective function of frequency support. In order to calculate $\Delta\omega_{max}$, the closed loop in Fig. 5 is opened [19], and the input frequency of the frequency modulation control is viewed as:

$$\Delta\omega_{in} = m_{\Delta\omega}t \tag{36}$$

where $m_{\Delta\omega}$ is the slope of the frequency intercept. According to the control structure in Fig. 5, the responses of the prime mover and DFIG can be obtained:

$$\Delta P_{Gi} = -\alpha_i m_{\Delta\omega}\left(t-T_i+T_i e^{-t/T_i}\right) \tag{37}$$

$$\Delta P_{DFIG} = m_{\Delta\omega}\left(\frac{A+B/r_1}{r_1-r_2}e^{r_1 t}+\frac{A+B/r_2}{r_2-r_1}e^{r_2 t}+\frac{B}{K_{i_pll}}\right) \tag{38}$$

By solving the non-linear equations in (39) formed by the conditions to obtain the maximum frequency deviation $d\Delta\omega/dt = 0$, $\Delta\omega_{max}$ can be calculated (see Appendix D for the derivation process)

$$\begin{cases} D\omega_{\max}+P_{step}+\sum_i^N\alpha_i m_{\Delta\omega}\left(t-T_i+T_i e^{-t/T_i}\right) \\ -m_{\Delta\omega}\left[\begin{array}{c}\frac{A}{r_1-r_2}e^{r_1 t}+\frac{A}{r_2-r_1}e^{r_2 t}+\frac{B}{K_{i_pll}} \\ +\frac{B}{r_1(r_1-r_2)}e^{r_1 t}+\frac{B}{r_2(r_2-r_1)}e^{r_2 t}\end{array}\right]=0 \\ \left[\begin{array}{c}\sum_i^N\left(\frac{\alpha_i T_i^2 m_{\Delta\omega}}{DT_i-2H}-\frac{2H\alpha_i m_{\Delta\omega}}{D^2}-\frac{\alpha_i T_i m_{\Delta\omega}}{D}\right)+\frac{P_{step}}{D} \\ -\frac{Bm_{\Delta\omega}}{DK_{i_pll}}-\frac{(A+B/r_2)2Hm_{\Delta\omega}}{(r_2-r_1)(D+2Hr_2)}-\frac{(A+B/r_1)2Hm_{\Delta\omega}}{(r_1-r_2)(D+2Hr_1)}\end{array}\right]e^{\frac{-Dt_{\max}}{2H}} \\ +\left(\frac{2Hm_{\Delta\omega}}{D^2}-\frac{m_{\Delta\omega}t_{\max}}{D}\right)\sum\alpha_i-\frac{m_{\Delta\omega}}{D}\sum\alpha_i T_i \\ -m_{\Delta\omega}\sum\frac{\alpha_i T_i^2}{DT_i-2H}e^{-t_{\max}/T_i}-\frac{P_{step}}{D}+\frac{(A+B/r_1)2Hm_{\Delta\omega}}{(r_1-r_2)(D+2Hr_1)}e^{r_1 t_{\max}} \\ +\frac{Bm_{\Delta\omega}}{DK_{i_pll}}+\frac{(A+B/r_2)2Hm_{\Delta\omega}}{(r_2-r_1)(D+2Hr_2)}e^{r_2 t_{\max}}-\Delta\omega_{\max}=0 \end{cases} \tag{39}$$

3.3. Parameter optimization based on Double-objective functions

According to (33), the objective function of oscillation suppression $J_1(K_1, K2)$ can be defined as:

Table 4
Parameters of the tested system.

Parameter	Symbol	Value	Parameter	Symbol	Value
Voltage level	LV	0.69 kV	Voltage level	MV	16.5 kV
Voltage level	Bus 39	345 kV	Capacity of DFIGs	S_N	1666 kVA
Nominal power of T1	S_{T1}	10,000 MVA	Nominal power of T2	S_{T2}	10,000MVA
Short-circuit voltage of T1	U_{k1}	6%	Short-circuit voltage of T2	U_{k2}	18.1%

Table 5
Different Disturbance Cases in Test System.

Case	Disturbance type	Location	Starting time	Ending time
1	3LG Fault	Line 4–14	5.1 s	5.2 s
2	Load fluctuation	Load 08	5.1 s	5.2 s
3*	2LG Fault	Line 4–14	5.1 s	5.2 s
4*	Voltage Ref Step Change		5.1 s	

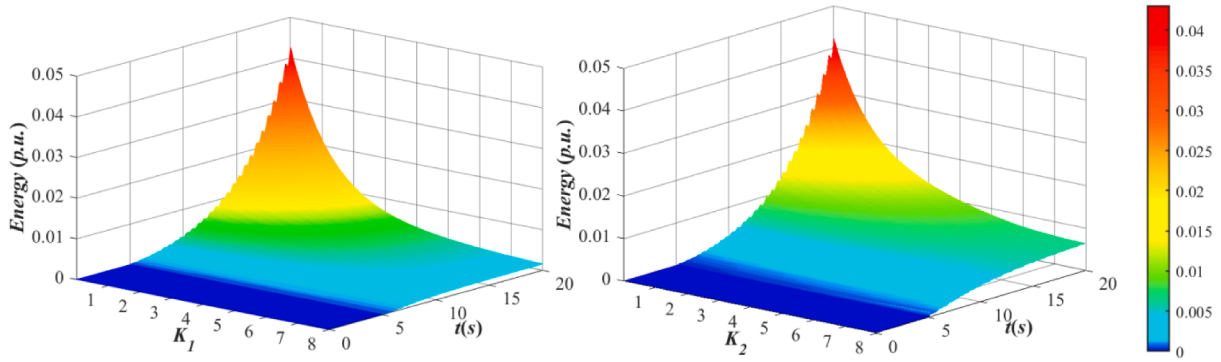
$$J_1(K_1, K_2) = \lg(\eta_1 + \eta_2 + 1) \tag{40}$$

In (40), η_1 and η_2 reflect the capability of branch 1 and branch 2 in providing positive energy curvature in the low frequency band. As η_1 and η_2 increase, if $J_1(K_1, K2)$ increases, the oscillation suppression capability of the compensation branches will improve. However, the energy curvature is affected by the oscillation mode. When the oscillation amplitude drops due to the suppression effect, the energy curvature in (29) will also decrease. In this case, the partial derivatives of the objective function with respect to the control parameters $\partial J_1 / \partial K_1$ and $\partial J_2 / \partial K_2$ should be above 0, and the 2nd-order partial derivatives $\partial^2 J_1 / \partial K_1^2$ and $\partial^2 J_2 / \partial K_2^2$ should be below 0. Thus, the logarithm of η_1 and η_2 are used in (40), which can reflect this trend more accurately.

With $\Delta\omega_{max}$ calculated from (39), the objective function of frequency support $J_2(K_1, K_2)$ can be defined as:

$$J_2(K_1, K_2) = \frac{\Delta\omega_{\max 0}}{\Delta\omega_{\max}(K_1, K_2)} \tag{41}$$

where $\Delta\omega_{\max 0}$ is the difference between the lowest frequency point and the rated frequency when there are no compensation branches. $\Delta\omega_{\max}(K_1, K_2)$ is the difference between the lowest frequency point and the rated frequency when the coefficients of the compensation branches are K_1 and K_2. This objective function can reflect the capability of the compensation branches in improving the lowest frequency point. The larger $J_2(K_1, K_2)$ is, the more the safety and stability of grid-connected



(a) Branch 1 (b) Branch 2.

Fig. 9. Variation of DFIG terminal energy with the parameters of compensation branches in Case 1.

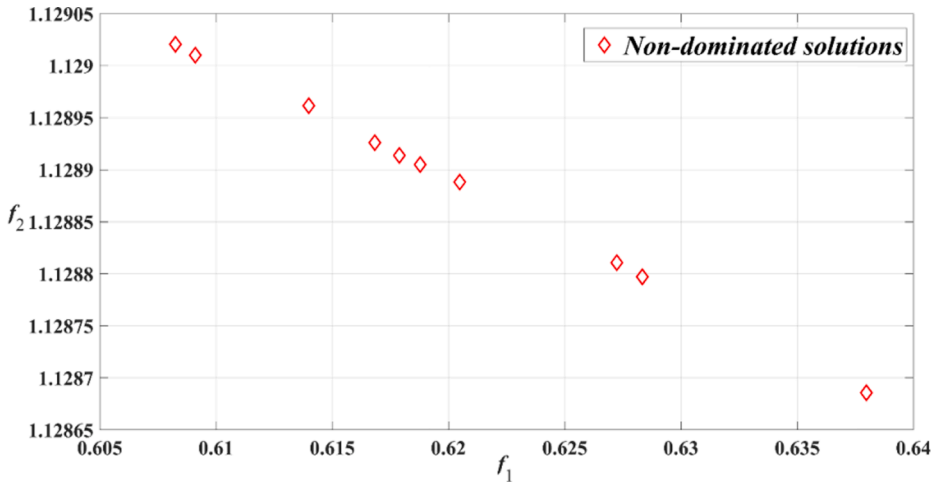


Fig. 10. Variation of DFIG terminal energy with the parameters of compensation branches in Case 2.

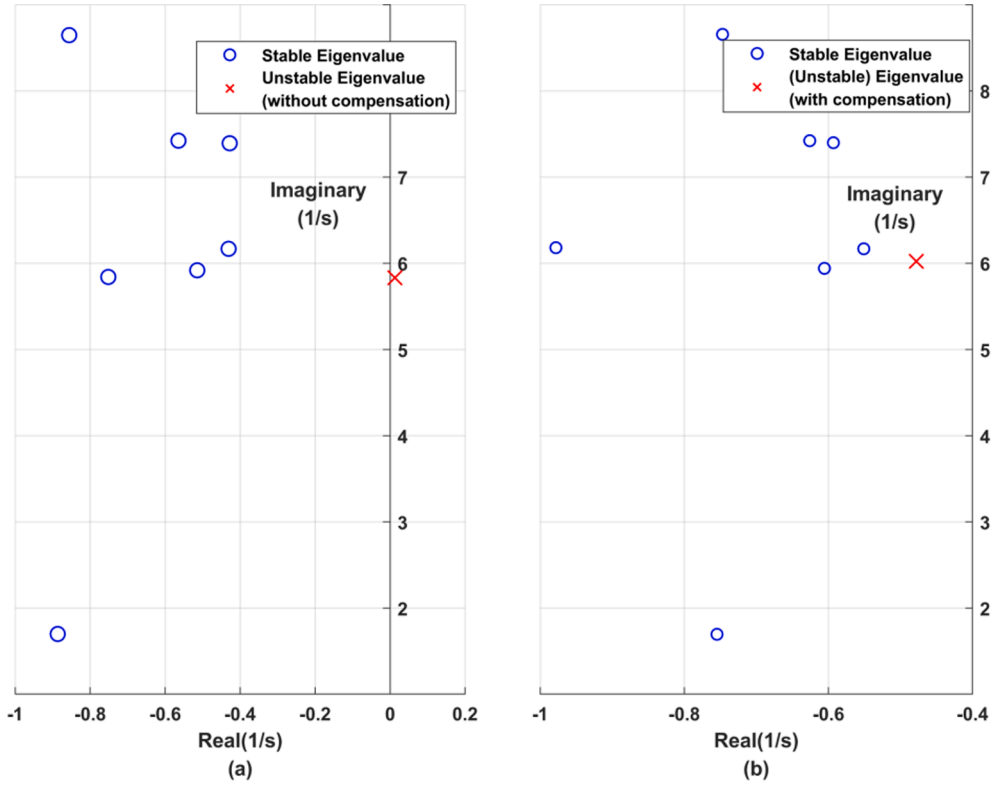


Fig. 11. Optimization solutions of MOGWO algorithm.

DFIG can be improved under the load impact.

Hence, the compensation branch parameter optimization model is *based* on double-objective functions of oscillation suppression and frequency support, constraints of system power flow and frequency lower limit, and the variables K_1 and K_2. The parameter optimization model is shown below:

$$\max\{J_1, J_2\} \quad s.t. \begin{cases} P(x) = 0 \\ \omega_{\min} \leqslant \omega \\ 0 \leqslant K_1, K_2 \leqslant K_{\max} \end{cases} \tag{42}$$

where $P(x) = 0$ is the constraint of power flow equations. $\omega_{\min}$ is the lower limit of system frequency. $0 \leq K_1$, $K_2 \leq K_{\max}$ ensures that each compensation branch provide positive energy curvature which does not exceed the upper limit of DFIG power regulation.

Concerning the above optimization model, first the multi-objective grey wolf optimization (MOGWO) algorithm is used to calculate the set of parameter solutions S. Then the optimal parameters (K^* 1, K^* 2) for the energy compensation branches are obtained by calculating the arithmetical mean values of the parameter solution set. The detailed calculation steps are as follows.

S1: Within the parameter search range, randomly generate the initial sampling set of parameters S_0.

S2: Apply the parameter set S_0 to (32) to calculate η_1 and η_2, and then apply η_1 and η_2 to the objective function of oscillation suppression J_1 in (40). Apply the parameter set S_0 to (39) to calculate $\Delta\omega_{max}$ and then apply $\Delta\omega_{max}$ to the objective function of frequency support J_2 in (41). Determine whether the constraints in (42) are satisfied or not. If yes, go to S3. if no, generate a new parameter set S_0 and repeat S2.

S3: According to the values of objective functions J_1 and J_2, use the MOGWO algorithm to update the parameter set by retaining the non-dominated parameter pairs.

S4: Repeat the search process in S2-S3 until reaching the maximum number of iterations. Calculate the arithmetical mean values of the present parameter set S, and the result is the optimal parameter pair (K^* 1, K^* 2) for the energy compensation branches.

4. Simulation verification

4.1. Test system

To verify the effectiveness and feasibility of the proposed method, the model of IEEE 10-machine 39-bus system connected with DFIG is built in RTDS. The hardware of generators applies real converters and controllers, which are connected with RTDS via GTAO card, GTDI card, AI and DO. The operation of power grid and disturbances are simulated in RTDS. The experimental platform is shown in Fig. 7. System parameters can be found in [20].

(a) RTDS (b) Experimental platform (c) Wiring of controller.

This paper substitutes synchronous generator G1 with 1000 DFIGs of 1.5 MW in shunt connection. As shown in Fig. 8, the DFIGs are first

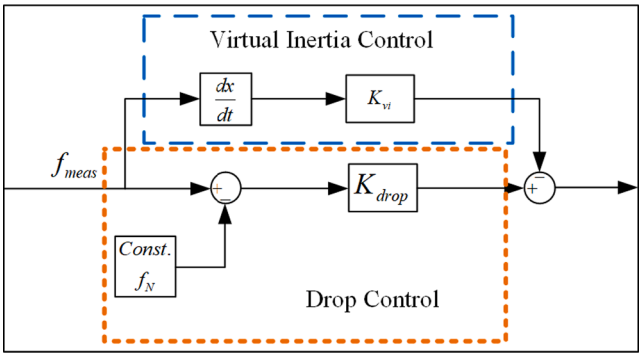


Fig. 12. Eigenvalues of the system. (a) Conventional control. (b) Compensation.

connected to the farm bus via a 0.69/20 kV transformer, and then connected to Bus 39 via a 20/230 kV transformer.

The main parameters of DFIG are shown in Table 1.

To verify the correctness and effectiveness of the proposed method, four disturbance cases are set to simulate the negative damping oscillation of DFIG-integrated system, i.e. transmission line fault (Case 1) and load fluctuation (Case 2). After the control parameters are optimized, the oscillation suppression effect and frequency support capability of the optimal energy compensation branches are validated. Besides, the active power, voltage and frequency of DFIG under 2LG fault (Case 3*) and step change of reference voltage (Case 4*) are depicted as shown in Fig. E-1 and E-2. The detailed settings of four cases are shown in Table 5.

4.2. Energy curvature compensation effect of supplementary branches

In order to verify the energy curvature compensation effect of the supplementary branches designed in Section 1.3, parameters K_1 and K_2 of branch 1 and branch 2 are increased while the other operating conditions of the DFIG-integrated system remain unchanged. The time-domain characteristics of DFIG terminal energy in Case 1 and Case 2 are recorded, as shown in Figs. 9 and 10 respectively.

(1) Case 1

Fig. 9(a) and (b) show the variation curves of DFIG terminal energy with the increasing of parameters K_1 and K_2 in Case 1. It can be seen that, when the compensation branches are not applied, the energy curvature is negative, and the energy reaches 3.293×10^{-1} p.u. at $t = 20$ s. The energy emitted by the system tends to increase, thus the oscillation will finally go unstable. According to Fig. 9(a), as K_1 increases, the energy curvature turns from negative to positive. When $K_1 = 8$, the energy stabilizes at 3.053×10^{-2} p.u. at $t = 20$ s. According to Fig. 9(b), as K_2 increases, the energy curvature turns form negative to positive. When $K_2 = 8$, the energy stabilizes at 7.478×10^{-2} p.u. at $t = 20$ s, and the system stops emitting energy.

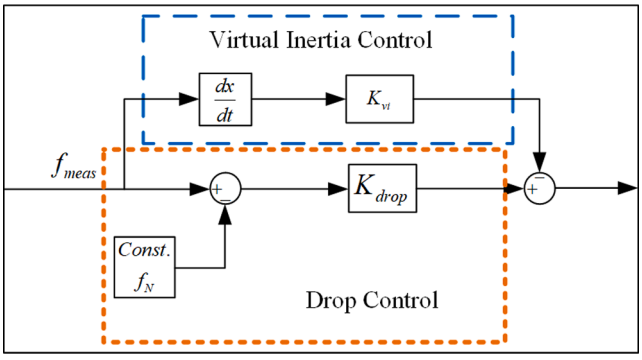


Fig. 13. Diagram of Virtual Inertia and Drop Control.

(2) Case 2

Fig. 10(a) and (b) show the variation curves of DFIG terminal energy with the increasing of parameters K_1 and K_2 in Case 2. It can be seen that, when the compensation branches are not applied, the energy curvature is negative, and the energy reaches 4.232×10^{-2} p.u. at $t = 20$ s. The energy emitted by the system tends to increase, thus the oscillation will finally go unstable. In Fig. 10(a), the energy curvature turns from negative to positive as K_1 increases. When $K_1 = 8$, the energy stabilizes at 1.595×10^{-3} p.u. at $t = 20$ s. In Fig. 10(b), the energy curvature turns form negative to positive as K_2 increases. When $K_2 = 8$, the energy stabilizes at 6.333×10^{-3} p.u. at $t = 20$ s, and the system stops emitting energy. According to the above simulation results, both branch 1 and branch 2 can provide positive energy curvature compensation in different disturbance cases. However, the compensation effect gradually decreases as the oscillation is suppressed.

4.3. Verification of parameter optimization effect

(3) Parameter optimization results

In order to verify the effectiveness of the double-objective functions proposed in Section 3.3, the parameter optimization model is built according to (41), and the MOGWO algorithm is used to calculate the set of parameter solutions. The calculation results are shown in Fig. 11, where the horizontal ordinate is objective function J_1, and the vertical ordinate is objective function J_2.

To balance objective functions J_1 and J_2, the arithmetical mean values of the parameter solutions are calculated, so that the optimal compensation coefficients are obtained: $K_1 = 2.5556$, $K_2 = 5.3189$.

The comparison between the system eigenvalues before and after the compensation branches are applied, as shown in Fig. 12.

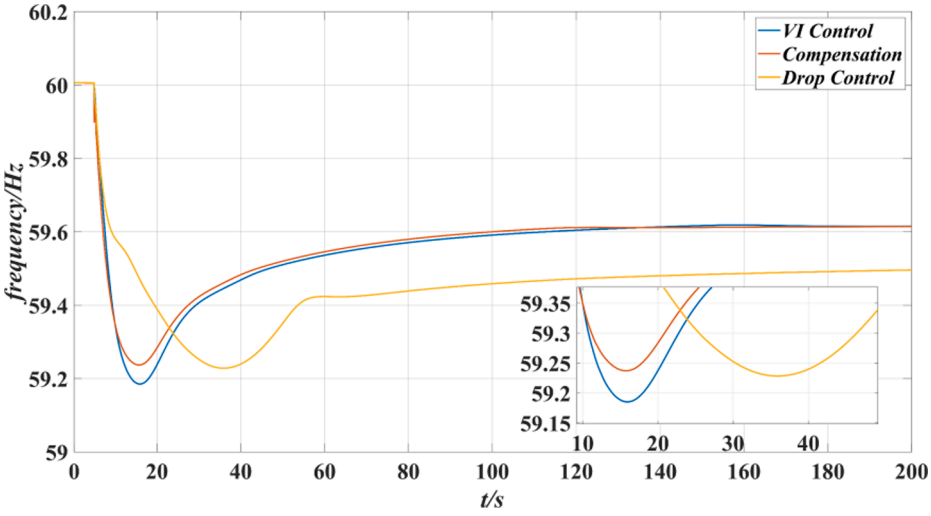


Fig. 14. Frequency responses of the system when different control strategies are applied.

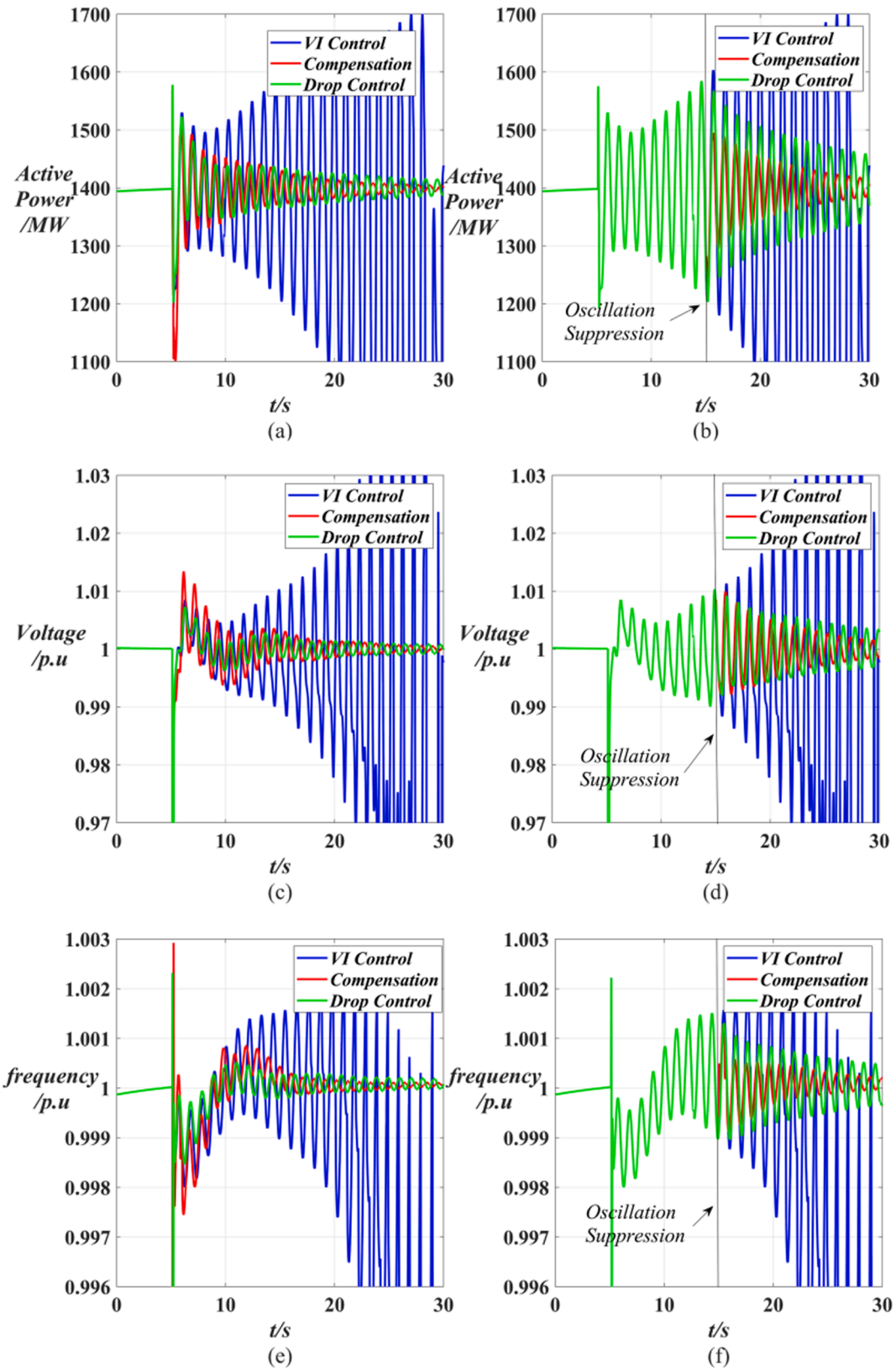


Fig. 15. Low frequency oscillation curves of DFIG in Case 1. (a) Active Power of DFIG when the control measure is added from the beginning. (b) Active Power of DFIG when the control measure is added 10 s after the oscillation occurs. (c) Voltage at DFIGs bus when the control measure is added from the beginning. (d) Voltage at DFIG bus when the control measure is added 10 s after the oscillation occurs. (e) Frequency at DFIG bus when the control measure is added from the beginning. (f) Frequency at DFIG bus when the control measure is added 10 s after the oscillation occurs.

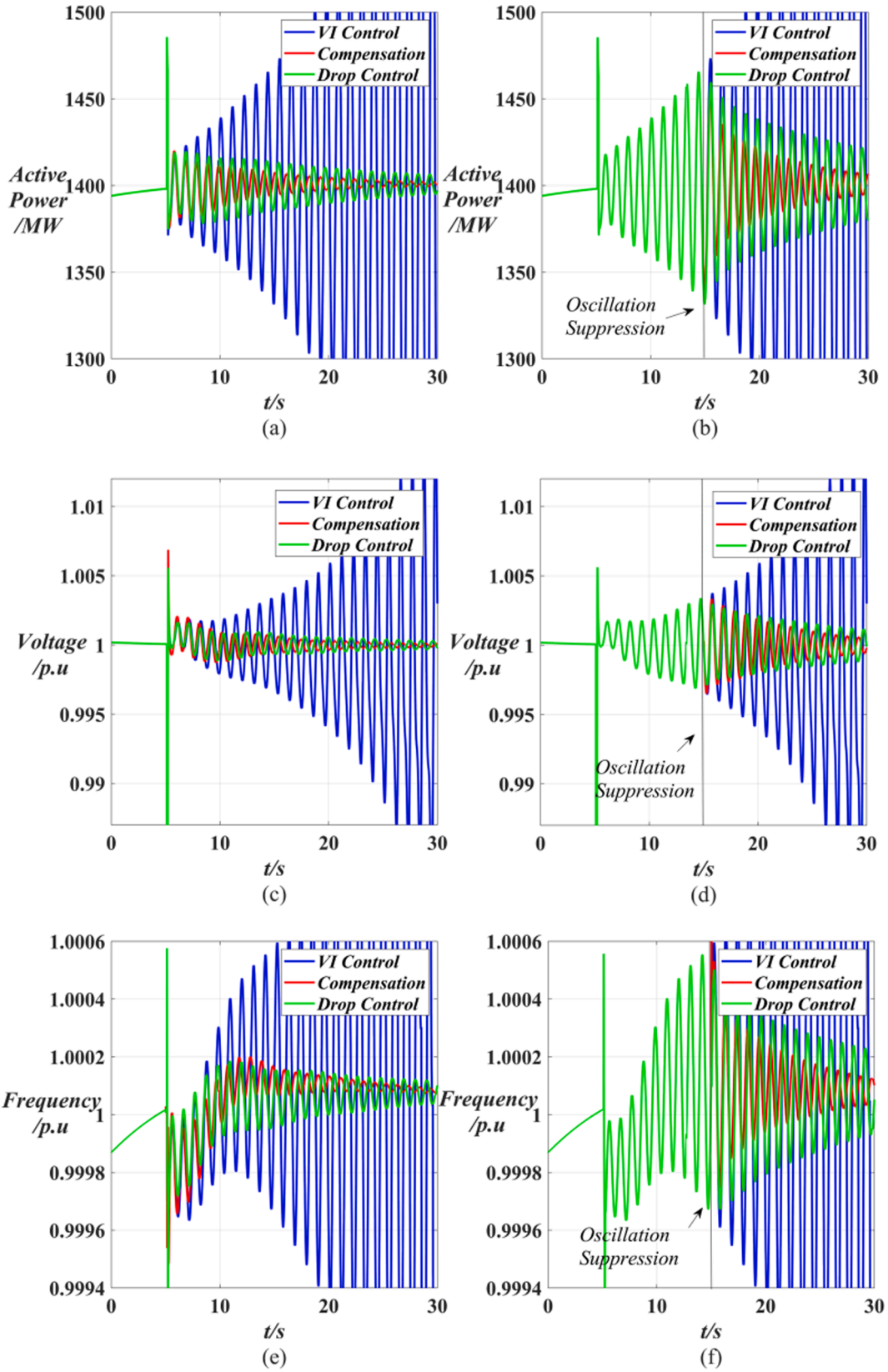


Fig. 16. Low frequency oscillation curves of DFIG in Case 2. **(a)** Active Power of DFIG when the control measure is added from the beginning. **(b)** Active Power of DFIG when the control measure is added 10 s after the oscillation occurs. **(c)** Voltage at DFIG bus when the control measure is added from the beginning. **(d)** Voltage at DFIG bus when the control measure is added 10 s after the oscillation occurs. **(e)** Frequency at DFIG bus when the control measure is added from the beginning. **(f)** Frequency at DFIG bus when the control measure is added 10 s after the oscillation occurs.

(4) Effect of parameter optimization in comparison with drop control

In order to verify whether the optimal compensation branches can balance oscillation suppression and frequency support, simulation tests are conducted to compare the response characteristics of low frequency oscillation in the following three scenarios: (1) when only the virtual inertia control is applied. (2) when the virtual inertia control and the optimal compensation branches are applied. (3) when the virtual inertia control and the droop control are applied (diagram shown in Fig. 13). In the three scenarios, the maximum frequency deviation remains the same.

Based on the original test system, the primary frequency modulation is simulated by increasing all the loads by 5% at $t = 5$ s. The droop control parameter K_D is set at 0.3. The frequency responses of the system in the above three scenarios are shown in Fig. 14.

As can be seen from Fig. 14, the compensation branches do not affect the steady-state frequency deviation, but affect the maximum frequency deviation. This is consistent with the theoretical analysis in Section 2.2. The optimal compensation branches can decrease the maximum frequency deviation during primary frequency modulation by about 6.4% (0.052 Hz), thus reducing the demand on active power dispatching from the other synchronous generators when system frequency recovers. In the scenario where the droop control is added, the steady-state frequency deviation is larger. This will cause the system to operate at a relatively low frequency, which is not good for the stability of system. Hence, it is verified that the optimal compensation branches can not only optimize the frequency support capability of DFIG, but also can improve the stability of low frequency oscillation. Besides, they are well compatible with the virtual inertia control.

Take the responses of generator G4 for example. Concerning the above three scenarios, the oscillation suppression effects of DFIG in Case 1 and Case 2 are shown in Figs. 15and 16 respectively.

Fig. 15(a), (c), (e) and (b), (d), (f) respectively show the oscillation suppression effects of three control measures in Case 1. At $t = 5.1$ s, a low frequency oscillation of 1.01 Hz occurs in the system due to the fault. When no control strategy is applied, the oscillation diverges to instability at $t = 29.8$ s. When the optimal compensation branches are applied from the beginning, the oscillation converges to stability within 20 s. When the droop control is applied, the oscillation also converges to stability, only at a lower converging speed. As shown in Fig. 15(b), (d) and (f), when the optimal compensation branches are applied 15 s before the instability, the oscillation is quickly suppressed within 15 s. The oscillation suppression speed of droop control is lower than that of the optimal branches.

Fig. 16(a), (c), (e) and (b), (d), (f) respectively show the oscillation suppression effects of three control measures in Case 2. At $t = 5.1$ s, a low frequency oscillation of 0.917 Hz occurs in the system due to the disturbance. When no control strategy is applied, the oscillation gradually diverges to instability. When the optimal compensation branches are applied from the beginning, the oscillation converges to stability within 20 s. When the droop control is applied, the oscillation also converges to stability, only at a lower converging speed. According to Fig. 16(b), (d) and (f), when the optimal compensation branches are applied 15 s before the instability, the oscillation is suppressed within 15 s. the oscillation suppression speed of droop control is lower than that of the optimal branches.

The proposed control strategy uses low-frequency measurements of 0.2–2.5 Hz as inputs. When DFIG is in normal operation state, the tracking effect has to do with the rotor angle control link and MPPT control link, etc., the time scales of which are much larger than that corresponding to 0.2–2.5 Hz, so the tracking accuracy is not affected. The major time delay of the proposed control strategy appears in the low-frequency component measuring link. According to reference [21], the existing low-frequency component measuring time is about 20 ms-70 ms. The proposed control strategy studies low frequency oscillations in the electromechanical transient, the time scale of which is a few seconds, so this time delay can meet the requirements of oscillation suppression.

5. Conclusion

This paper proposes a method based on dynamic energy curvature for suppressing low frequency oscillations of DFIG with virtual inertia control. The main conclusions are as follows:

(1) A method to improve the system stability is proposed by compensating for the energy curvature. The sign (positive or negative) of the energy curvature directly reflects the stability of the system power. Thus, unlike traditional stability control methods which choose the measurements randomly, this method uses the measurements that can compensate for the negative energy curvature as the input of stability control. The control structure is simple, and the control principle is clear.
(2) The supplementary energy branches can effectively improve the curvature of DFIG terminal energy, reduce the energy of the system, improve the system's ability to withstand oscillation, and inhibit the development of low frequency oscillation. The compensation branches can achieve rapid suppression within a few seconds after different types of disturbances to the system. This prevents system instability that would result from excessive oscillation amplitude, thereby improving the system's rotor angle stability.
(3) This paper establishes a parameter optimization scheme based on the double-objective functions of oscillation suppression and frequency support. The energy compensation branches after the optimized parameter configuration can improve oscillation suppression while also improving the frequency support capability of DFIG during the primary frequency modulation. This prevents the system from exceeding the frequency tolerance range and helps improve the frequency stability of the system.
(4) The proposed optimization strategy does not require real-time computation of the system matrix, so the calculation time is greatly reduced. This makes it possible to adjust the compensation branch parameters according to the variation of the equilibrium point.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A

According to Eqs. (6) and (14), $\Delta\theta_{pll}$, $\Delta\theta_s$ and $\Delta\theta_{pcc}$ have the same frequency. It may be assumed that:

$$\Delta\theta_{pcc} - \Delta\theta_{pll} = \Omega e^{\gamma t}\sin(\omega t + \theta) \tag{A-1}$$

Applying (A-1) to (14) yields:

$$\begin{aligned} &\Delta\theta_s - \Delta\theta_{pll} = \Omega_U e^{\gamma t}\sin(\omega t + \theta) \\ &\Omega_U = \frac{U_{pcc}}{U_s}\frac{\sin(\theta_Z + \theta_{s.0} - \varphi - \theta_{pcc.0})}{\sin(\theta_Z - \varphi)}\Omega \end{aligned} \tag{A-2}$$

Eq. (13) can be linearized as:

$$\begin{aligned} &U_s\cos(\theta_{s.0} - \theta_{pll.0})(\Delta\theta_s - \Delta\theta_{pll}) \\ &= U_{pcc}\cos(\theta_{pcc.0} - \theta_{pll.0})(\Delta\theta_{pcc} - \Delta\theta_{pll}) \\ &+ZI_s\cos(\theta_Z + \theta_{s.0} - \theta_{pll.0} + \varphi)(\Delta\theta_s - \Delta\theta_{pll} - \Delta\varphi) \end{aligned} \tag{A-3}$$

Thus applying (A-1) and (A-2) to (A-3) yields:

$$\begin{aligned} &\Delta\theta_s - \Delta\theta_{pll} - \Delta\varphi \\ &= \frac{U_{pcc}\begin{bmatrix}\cos(\theta_{s.0} - \theta_{pll.0}) \\ \cdot\sin(\theta_Z + \theta_{s.0} - \varphi - \theta_{pcc.0}) \\ -\cos(\theta_{pcc.0} - \theta_{pll.0})\sin(\theta_Z - \varphi)\end{bmatrix}}{ZI_s\sin(\theta_Z - \varphi)\cos(\theta_Z + \theta_{s.0} - \theta_{pll.0} - \varphi)}\Omega e^{\gamma t}\sin(\omega t + \theta) \\ &= \frac{U_{pcc}\begin{bmatrix}\sin(\theta_Z - \varphi + 2\theta_{s.0} - \theta_{pll.0} - \theta_{pcc.0}) \\ -\sin(\theta_Z + \theta_{pcc.0} - \varphi - \theta_{pll.0})\end{bmatrix}}{2ZI_s\sin(\theta_Z - \varphi)\cos(\theta_Z + \theta_{s.0} - \theta_{pll.0} - \varphi)}\Omega e^{\gamma t}\sin(\omega t + \theta) \\ &= \Omega_I e^{\gamma t}\sin(\omega t + \theta) \end{aligned} \tag{A-4}$$

Appendix B

According to Fig. 2, the rotor current response can be expressed as:

$$\begin{cases} \Delta i_{rd} = K_p(\Delta P_{ref} - \Delta P) + K_i\int(\Delta P_{ref} - \Delta P)dt \\ \Delta i_{rq} = K_p(\Delta Q_{ref} - \Delta Q) + K_i\int(\Delta Q_{ref} - \Delta Q)dt \end{cases} \tag{B-1}$$

Applying (6) and (18) to (B-1) yields:

$$\begin{cases} \Delta i_{rd} = -K_p(U_{sd0}\Delta i_{sd} + I_{sd0}\Delta u_{sd} + I_{sq0}\Delta u_{sq}) \\ \quad -K_i\int(U_{sd0}\Delta i_{sd} + I_{sd0}\Delta u_{sd} + I_{sq0}\Delta u_{sq})dt \\ \quad -K_{VI}K_pK_{p_pll}\dfrac{d\Delta u_{sq}}{dt} - K_{VI}K_{i_pll}K_p\Delta u_{sq} \\ \quad -K_{VI}K_{p_pll}K_i\Delta u_{sq} - K_{VI}K_{i_pll}K_i\int\Delta u_{sq}dt \\ \Delta i_{rq} = -K_p(I_{sd0}\Delta u_{sq} - I_{sq0}\Delta u_{sd} - U_{sd0}\Delta i_{sq}) \\ \quad -K_i\int(I_{sd0}\Delta u_{sq} - I_{sq0}\Delta u_{sd} - U_{sd0}\Delta i_{sq})dt \end{cases} \tag{B-2}$$

By applying (12) to (B-2), Δi_{rd} and Δi_{rq} can be calculated:

$$\begin{cases} \Delta i_{rd} = A_d e^{\gamma t}\sin(\omega t + \theta) + B_d e^{\gamma t}\cos(\omega t + \theta) \\ \Delta i_{rq} = A_q e^{\gamma t}\sin(\omega t + \theta) + B_q e^{\gamma t}\cos(\omega t + \theta) \end{cases} \tag{B-3}$$

where,

$$\begin{cases} A_d = \begin{bmatrix} \Omega_I U_{sd0} I_s \sin(\theta_{s.0} - \theta_{pll.0} + \varphi) \\ +\Omega_U U_s I_{sd0} \sin(\theta_{s.0} - \theta_{pll.0}) \\ -\Omega_U U_s I_{sq0} \cos(\theta_{s.0} - \theta_{pll.0}) \end{bmatrix} \left(K_p + \dfrac{K_i \gamma}{\gamma^2 + \omega^2} \right) \\ -\Omega_U U_s \cos(\theta_{s.0} - \theta_{pll.0}) K_{VI} \begin{pmatrix} K_{p_pll} K_p \gamma + K_{i_pll} K_p \\ +K_{p_pll} K_i + K_{i_pll} K_i \dfrac{\gamma}{\gamma^2 + \omega^2} \end{pmatrix} \\ A_q = \dfrac{K_i \omega}{\gamma^2 + \omega^2} \begin{bmatrix} \Omega_U U_s I_{sq0} \cos(\theta_{s.0} - \theta_{pll.0}) \\ -\Omega_I U_{sd0} I_s \sin(\theta_{s.0} - \theta_{pll.0} + \varphi) \\ -\Omega_U U_s I_{sd0} \sin(\theta_{s.0} - \theta_{pll.0}) \end{bmatrix} \end{cases}$$

$$\begin{cases} B_d = -\begin{bmatrix} \Omega_I U_{sd0} I_s \sin(\theta_{s.0} - \theta_{pll.0} + \varphi) \\ +\Omega_U U_s I_{sd0} \sin(\theta_{s.0} - \theta_{pll.0}) \\ -\Omega_U U_s I_{sq0} \cos(\theta_{s.0} - \theta_{pll.0}) \end{bmatrix} K_i \dfrac{\omega}{\gamma^2 + \omega^2} \\ -\Omega_U U_s \cos(\theta_{s.0} - \theta_{pll.0}) K_{VI} \left(K_{p_pll} K_p \omega - K_{i_pll} K_i \dfrac{\omega}{\gamma^2 + \omega^2} \right) \\ B_q = \dfrac{K_i \omega}{\gamma^2 + \omega^2} \left\{ \begin{array}{c} \dfrac{1}{2} \Omega_U U_s I_{sd0} \begin{bmatrix} \cos(\theta_{s.0} - \theta_{pll.0} + \varphi) \\ -\cos(\theta_{s.0} - \theta_{pll.0} - \varphi) \end{bmatrix} \\ +\Omega_U U_s I_{sq0} \sin(\theta_{s.0} - \theta_{pll.0}) \end{array} \right\} \end{cases}$$

Appendix C

G_{DFIG}(s) is mainly affected by PLL, virtual inertia and the compensation branches:

$$\Delta P_e(\Delta\omega_s) = \Delta P_{VI} + \Delta P_{EC} \tag{C-1}$$

where,

$$\Delta P_{VI} = -K_{VI} \frac{K_{p_pll} U_s s^2 + K_{i_pll} U_s s}{s^2 + K_{p_pll} U_s s + K_{i_pll} U_s} \Delta\omega_s \tag{C-2}$$

$$\Delta P_{EC} = U_s \Delta i'_{sd} = \frac{U_s \begin{bmatrix} K_1 U_s \cos(\theta_{s.0} - \theta_{pll.0}) \\ -K_2 I_s \sin(\theta_{s.0} - \theta_{pll.0} - \varphi) \end{bmatrix} s}{s^2 + K_{p_pll} U_s s + K_{i_pll} U_s} \Delta\omega_s \tag{C-3}$$

By combining (C-2) and (C-3), the transfer function of DFIG primary frequency modulation response G_{DFIG}(s) can be obtained:

$$\begin{aligned} G_{DFIG}(s) = & -\frac{K_{VI} K_{p_pll} s^2}{s^2 + K_{p_pll} s + K_{i_pll}} \Delta\omega_s \\ & -\frac{\begin{bmatrix} K_{VI} K_{i_pll} - K_1 U_s \cos(\theta_{s.0} - \theta_{pll.0}) \\ +K_2 I_s \sin(\theta_{s.0} - \theta_{pll.0} - \varphi) \end{bmatrix} s}{s^2 + K_{p_pll} s + K_{i_pll}} \Delta\omega_s \end{aligned} \tag{C-4}$$

Appendix D

The open-loop response model of the primary frequency modulation is:

$$\frac{\Delta\omega}{\Delta P} = \frac{1}{2Hs + D + K_{SG} \sum_i G_{GOVi}(s) + K_{DFIG} G_{DFIG}(s)} \tag{D-1}$$

The 1st-order equivalent model of the governor of G_i is:

$$G_{GOVi} = \frac{-\alpha_i}{1+sT_i} \tag{D-2}$$

The equivalent model of DFIG is:

$$G_{DFIG}(s) = -\frac{K_{VI}K_{p_pll}s^2}{s^2+K_{p_pll}s+K_{i_pll}}\Delta\omega_s - \frac{\begin{bmatrix} K_{VI}K_{i_pll} - K_1U_s\cos(\theta_{s.0}-\theta_{pll.0}) \\ +K_2I_s\sin(\theta_{s.0}-\theta_{pll.0}-\varphi)\end{bmatrix}s}{s^2+K_{p_pll}s+K_{i_pll}}\Delta\omega_s \tag{D-3}$$

By applying (38) to (D-2), the prime mover response can be calculated:

$$\Delta P_{Gi} = -\alpha_i m_{\Delta\omega}\left(t - T_i + T_i e^{-t/T_i}\right) \tag{D-4}$$

By applying (36) to (D-3), the DFIG response can be calculated:

$$\Delta P_{DFIG} = m_{\Delta\omega}\begin{bmatrix} \frac{A}{r_1-r_2}e^{r_1t} + \frac{A}{r_2-r_1}e^{r_2t} + \frac{B}{K_{i_pll}} \\ +\frac{B}{r_1(r_1-r_2)}e^{r_1t} + \frac{B}{r_2(r_2-r_1)}e^{r_2t}\end{bmatrix} \tag{D-5}$$

where,

$$A = -K_{DFIG}K_{VI}K_{p_pll}, B = -K_{DFIG}\begin{bmatrix} K_{VI}K_{i_pll} - K_1U_s\cos(\theta_{s.0}-\theta_{pll.0}) \\ +K_2I_s\sin(\theta_{s.0}-\theta_{pll.0}-\varphi)\end{bmatrix}$$

$$r_1 = \frac{-K_{p_pll}+\sqrt{K_{p_pll}^2-4K_{i_pll}}}{2}, r_2 = \frac{-K_{p_pll}-\sqrt{K_{p_pll}^2-4K_{i_pll}}}{2}$$

According to the open-loop response model:

$$2H\frac{d\Delta\omega}{dt} + D\Delta\omega = -P_{step} + K_{SG}\sum_i^N \Delta P_{GI} + K_{DFIG}\Delta P_{DFIG} \tag{D-6}$$

Suppose the maximum frequency deviation $\Delta\omega_{\max}$ appears at $t = t_{\max}$. Applying (D-4) and (D-5) to (D-6) yields:

$$\begin{aligned}\Delta\omega_{\max} = &\begin{bmatrix} \sum_i^N\left(\frac{\alpha_iT_i^2m_{\Delta\omega}}{DT_i-2H} - \frac{2H\alpha_im_{\Delta\omega}}{D^2} - \frac{\alpha_iT_im_{\Delta\omega}}{D}\right) + \frac{P_{step}}{D} \\ -\frac{Bm_{\Delta\omega}}{DK_{i_pll}} - \frac{(A+B/r_2)2Hm_{\Delta\omega}}{(r_2-r_1)(D+2Hr_2)} - \frac{(A+B/r_1)2Hm_{\Delta\omega}}{(r_1-r_2)(D+2Hr_1)}\end{bmatrix}e^{\frac{-Dt_{\max}}{2H}} \\ &+\left(\frac{2Hm_{\Delta\omega}}{D^2} - \frac{m_{\Delta\omega}t_{\max}}{D}\right)\sum\alpha_i - \frac{m_{\Delta\omega}}{D}\sum\alpha_iT_i \\ -m_{\Delta\omega}\sum\frac{\alpha_iT_i^2}{DT_i-2H}e^{-t_{\max}/T_i} &- \frac{P_{step}}{D} + \frac{(A+B/r_1)2Hm_{\Delta\omega}}{(r_1-r_2)(D+2Hr_1)}e^{r_1t_{\max}} \\ &+\frac{Bm_{\Delta\omega}}{DK_{i_pll}} + \frac{(A+B/r_2)2Hm_{\Delta\omega}}{(r_2-r_1)(D+2Hr_2)}e^{r_2t_{\max}}\end{aligned} \tag{D-7}$$

When the maximum frequency deviation appears, $d\Delta\omega/dt = 0$. Thus according to (D-3)-(D-5):

$$\begin{aligned}D\omega_{\max} = &-P_{step} - K_{SG}\sum_i^N\alpha_im_{\Delta\omega}\left(t - T_i + T_ie^{-t/T_i}\right) \\ &+K_{DFIG}m_{\Delta\omega}\begin{bmatrix} \frac{A}{r_1-r_2}e^{r_1t} + \frac{A}{r_2-r_1}e^{r_2t} + \frac{B}{K_{i_pll}} \\ +\frac{B}{r_1(r_1-r_2)}e^{r_1t} + \frac{B}{r_2(r_2-r_1)}e^{r_2t}\end{bmatrix}\end{aligned} \tag{D-8}$$

Then $\Delta\omega_{\max}$ can be calculated according to (D-7) and (D-8) using the Newton-Raphson method.

Appendix E

Figs. E1 and E2

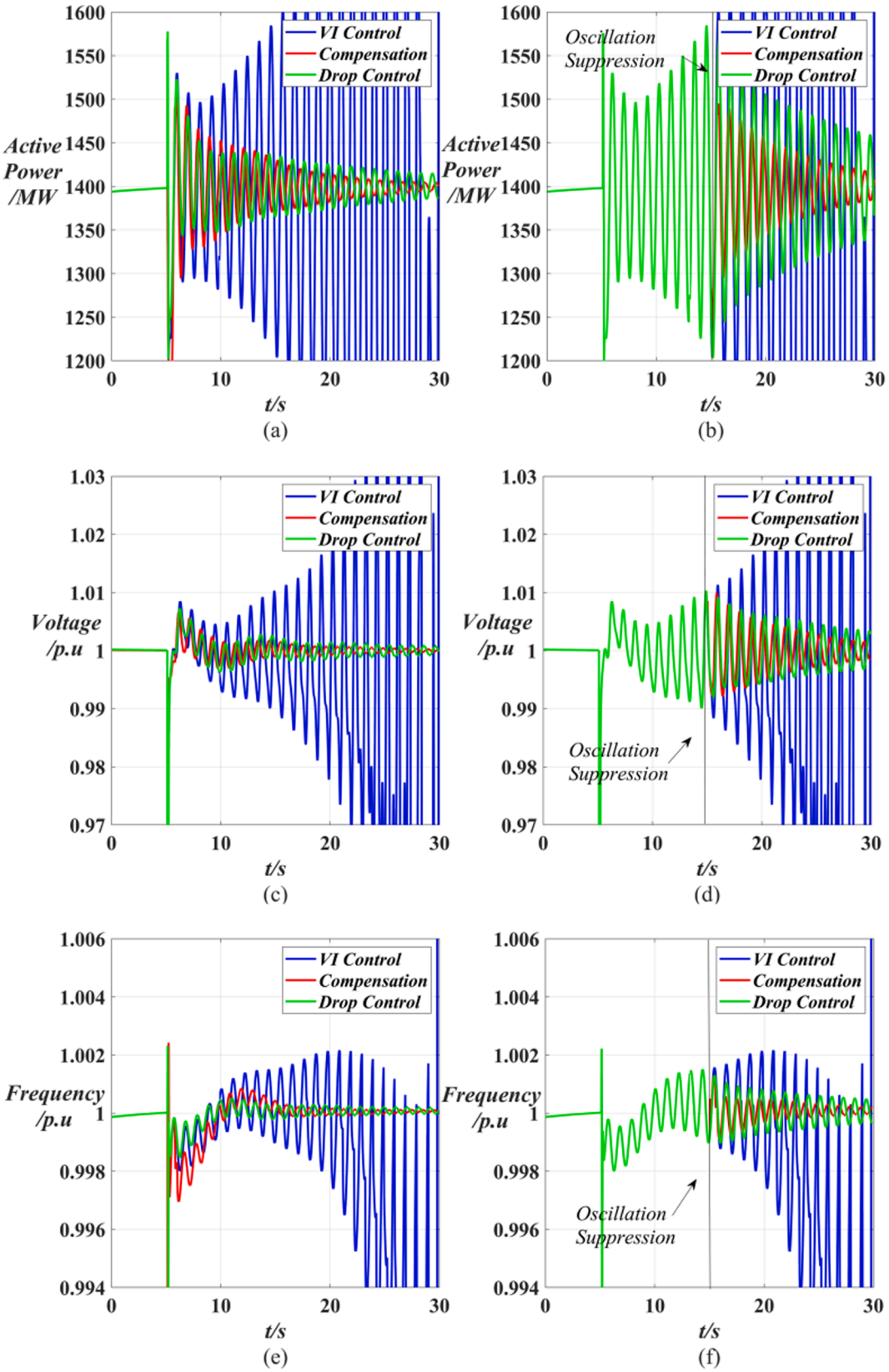


Fig. E1. Low frequency oscillation curves of DFIG in Case 3. (a) Active Power of DFIG when the control measure is added from the beginning. (b) Active Power of DFIG when the control measure is added 10 s after the oscillation occurs. (c) Voltage at DFIG bus when the control measure is added from the beginning. (d) Voltage at DFIG bus when the control measure is added 10 s after the oscillation occurs. (e) Frequency at DFIG bus when the control measure is added from the beginning. (f) Frequency at DFIG bus when the control measure is added 10 s after the oscillation occurs.

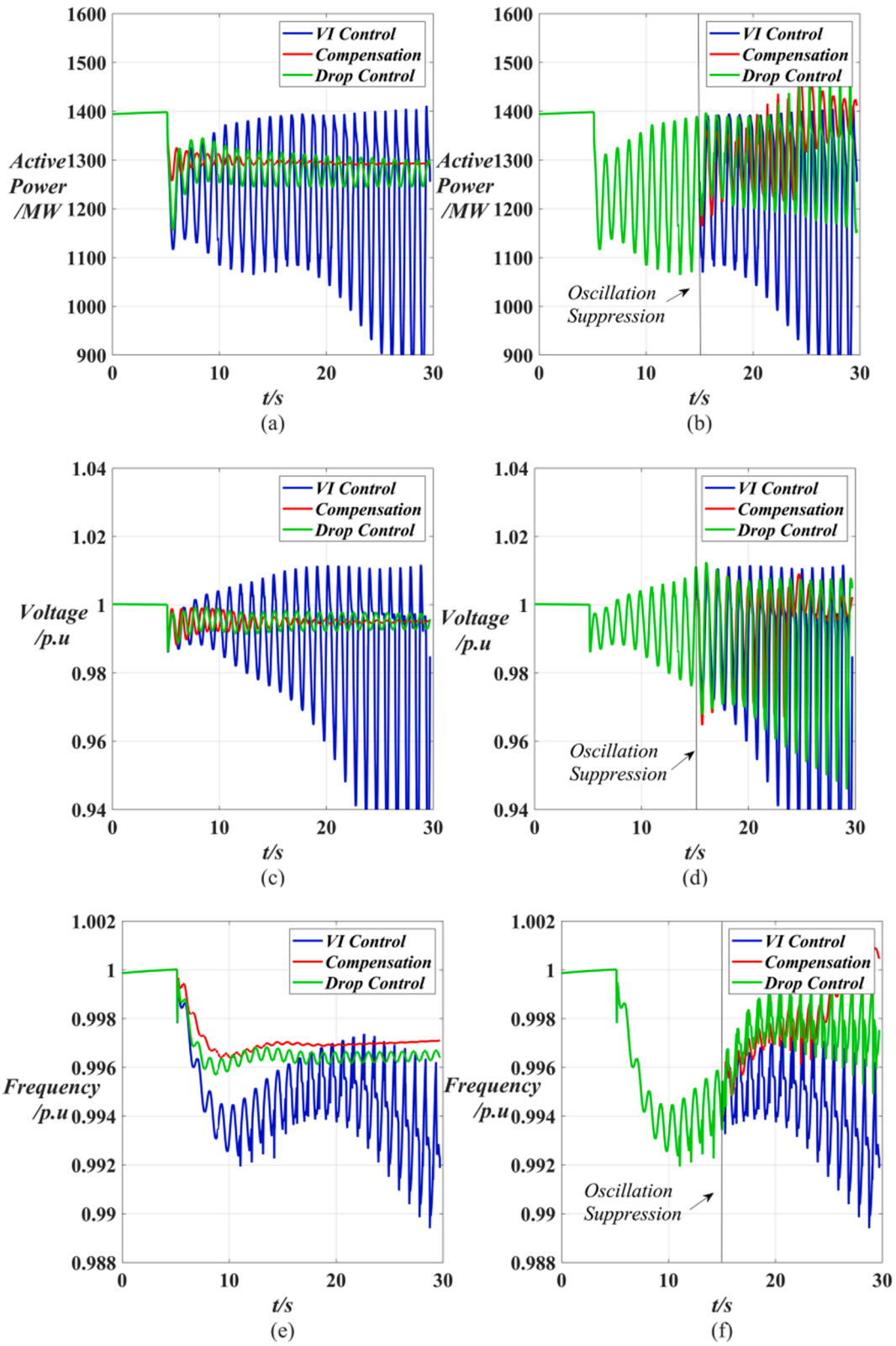


Fig. E2. Low frequency oscillation curves of DFIG in Case 4. (a) Active Power of DFIG when the control measure is added from the beginning. (b) Active Power of DFIG when the control measure is added 10 s after the oscillation occurs. (c) Voltage at DFIGs bus when the control measure is added from the beginning. (d) Voltage at DFIG bus when the control measure is added 10 s after the oscillation occurs. (e) Frequency at DFIG bus when the control measure is added from the beginning. (f) Frequency at DFIG bus when the control measure is added 10 s after the oscillation occurs.

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